

Chapter Two

2. Numerical Methods for IVPs

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2.1 Numerical Methods for First Order IVPs

Several numerical methods exist to approximate the solution of an IVP. Some common methods include: Euler Method, Runge-Kutta Methods, and so on.

2.1.1 Euler's Method

In this section, we introduce **Euler's method**, the most elementary numerical technique for solving initial-value problems (IVPs). Although it is seldom used in practice due to its low accuracy, its simplicity makes it an excellent tool for illustrating the principles behind more advanced methods.

The objective is to approximate the solution to the well-posed initial-value problem:

$$\begin{cases} \frac{dy}{dt} = f(t, y), & a \leq t \leq b, \\ y(a) = \alpha. \end{cases}$$

We do not obtain a continuous approximation to $y(t)$; instead, we compute approximations at discrete points $t_0, t_1, \dots, t_N$ in the interval $[a, b]$, called **grid points**.

Divide $[a, b]$ into N equal subintervals with step size:

$$h = \frac{b - a}{N}.$$

The grid points are:

$$t_i = a + ih, \quad i = 0, 1, 2, \dots, N.$$

Assume the true solution $y(t)$ has two continuous derivatives on $[a, b]$. By Taylor's Theorem, for each $i = 0, 1, \dots, N - 1$, there exists $\zeta_i \in (t_i, t_{i+1})$ such that:

$$y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\zeta_i).$$

Since $y'(t_i) = f(t_i, y(t_i))$, we have:

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + T_i,$$

where $T_i = \frac{h^2}{2}y''(\zeta_i)$ is the **local truncation error**, which is $O(h^2)$.

Euler's method constructs an approximation $w_i \approx y(t_i)$ by discarding the error term. The recurrence is:

$$\begin{cases} w_0 = \alpha, \\ w_{i+1} = w_i + hf(t_i, w_i), \quad i = 0, 1, \dots, N-1. \end{cases}$$

Algorithm: Explicit Euler Method

Input:

- $f(t, y)$: RHS of ODE $y' = f(t, y)$
- a, b : interval endpoints
- N : number of steps
- α : initial value $y(a) = \alpha$

Output:

Sequence (t_i, w_i) , $i = 0, \dots, N$, approximating $y(t_i)$

1. $h \leftarrow \frac{b-a}{N}$
2. $t_0 \leftarrow a$
3. $w_0 \leftarrow \alpha$
4. Output (t_0, w_0)
5. **for** $i = 0$ to $N-1$ **do**
 - $w_{i+1} \leftarrow w_i + h \cdot f(t_i, w_i)$ (Euler update)
 - $t_{i+1} \leftarrow t_i + h$ (or equivalently $a + (i+1)h$)
 - Output (t_{i+1}, w_{i+1})
6. **end for**

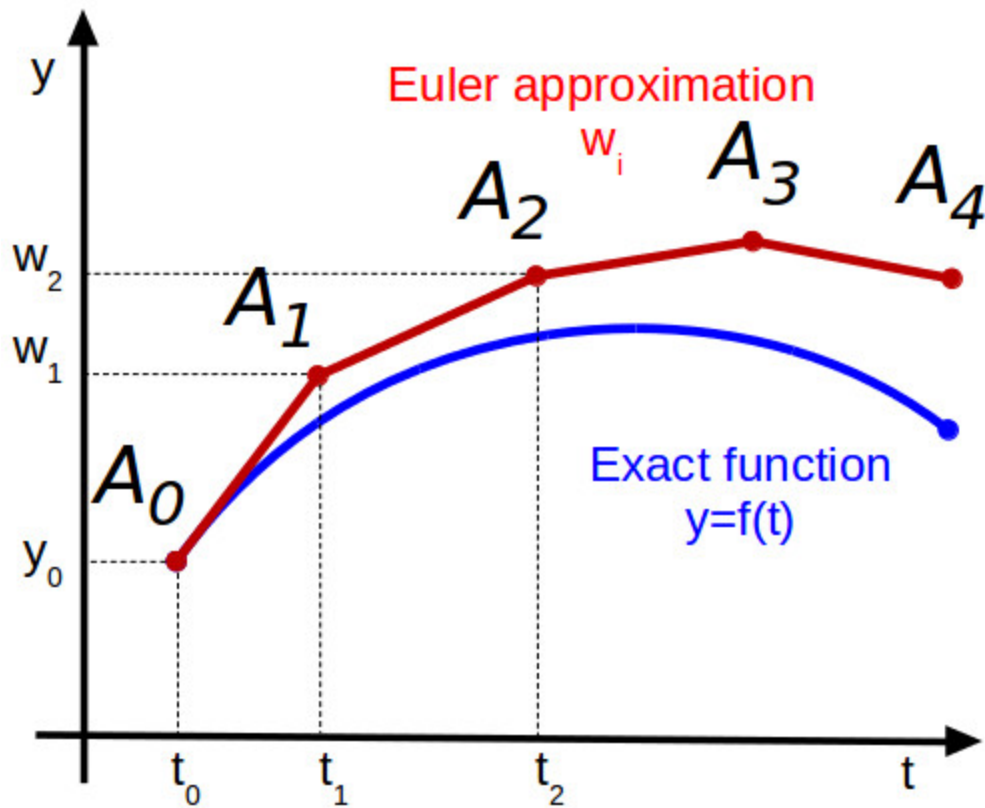


Figure: Geometric interpretation of Euler's method: the slope at each point is used to step forward.

Example:

Use Euler's method with $h = 0.2$ to solve the IVP:

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5.$$

The exact solution is:

$$y(t) = (t + 1)^2 - 0.5e^t.$$

Solution:

Here, $f(t, y) = y - t^2 + 1$, $a = 0$, $b = 2$, $h = 0.2$, so $N = \frac{2-0}{0.2} = 10$.

We compute iteratively using

$$w_{i+1} = w_i + h \cdot f(t_i, w_i), \quad w_0 = 0.5.$$

$$\begin{aligned}
t_0 = 0.0: \quad w_0 &= 0.5 \\
t_1 = 0.2: \quad w_1 &= 0.5 + 0.2(0.5 - 0^2 + 1) = 0.5 + 0.2(1.5) = 0.8 \\
t_2 = 0.4: \quad w_2 &= 0.8 + 0.2(0.8 - 0.04 + 1) = 0.8 + 0.2(1.76) = 1.152 \\
t_3 = 0.6: \quad w_3 &= 1.152 + 0.2(1.152 - 0.16 + 1) = 1.5504 \\
t_4 = 0.8: \quad w_4 &= 1.5504 + 0.2(1.5504 - 0.36 + 1) = 1.98848 \\
t_5 = 1.0: \quad w_5 &= 1.98848 + 0.2(1.98848 - 0.64 + 1) = 2.458176 \\
t_6 = 1.2: \quad w_6 &= 2.458176 + 0.2(2.458176 - 1.00 + 1) = 2.9498112 \\
t_7 = 1.4: \quad w_7 &= 2.9498112 + 0.2(2.9498112 - 1.44 + 1) = 3.45177344 \\
t_8 = 1.6: \quad w_8 &= 3.45177344 + 0.2(3.45177344 - 1.96 + 1) = 4.069128128 \\
t_9 = 1.8: \quad w_9 &= 4.069128128 + 0.2(4.069128128 - 2.56 + 1) = 4.7629537536 \\
t_{10} = 2.0: \quad w_{10} &= 4.7629537536 + 0.2(4.7629537536 - 3.24 + 1) = \mathbf{5.26754450432}
\end{aligned}$$

Thus, the Euler approximation at $t = 2$ is:

$$w_{10} \approx 5.2675.$$

The exact value is:

$$y(2) = (2 + 1)^2 - 0.5e^2 = 9 - 0.5 \cdot e^2 \approx 9 - 0.5 \cdot 7.389056 = 9 - 3.694528 = 5.305472.$$

Absolute error:

$$|y(2) - w_{10}| \approx |5.3055 - 5.2675| = 0.0380.$$

Note: Euler's method has

- **Local truncation error:** $O(h^2)$,
- **Global truncation error:** $O(h)$.

Hence, halving h roughly halves the global error. While simple, its first-order accuracy limits practical use — higher-order methods (e.g., RK4) are preferred for efficiency and precision.

```

In [1]: def euler_method(fun, a, b, y0, N):
import numpy as np

h = (b - a) / N
t = np.zeros(N + 1)
y = np.zeros(N + 1)

t[0] = a
y[0] = y0

for i in range(N):
    t[i + 1] = t[i] + h
    y[i + 1] = y[i] + h * fun(t[i], y[i])

return t, y

```

```

In [2]: import numpy as np
import matplotlib.pyplot as plt

# Define the ODE RHS and exact solution
def fun(t, y):
    return y - t**2 + 1

def exact_y(t):
    return (t + 1)**2 - 0.5 * np.exp(t)

# Parameters
y0 = 0.5      # initial condition
a = 0.0       # start time
b = 2.0       # end time
N = 10        # number of steps → h = 0.2

# Solve using Euler's method
t, y_num = euler_method(fun, a, b, y0, N)

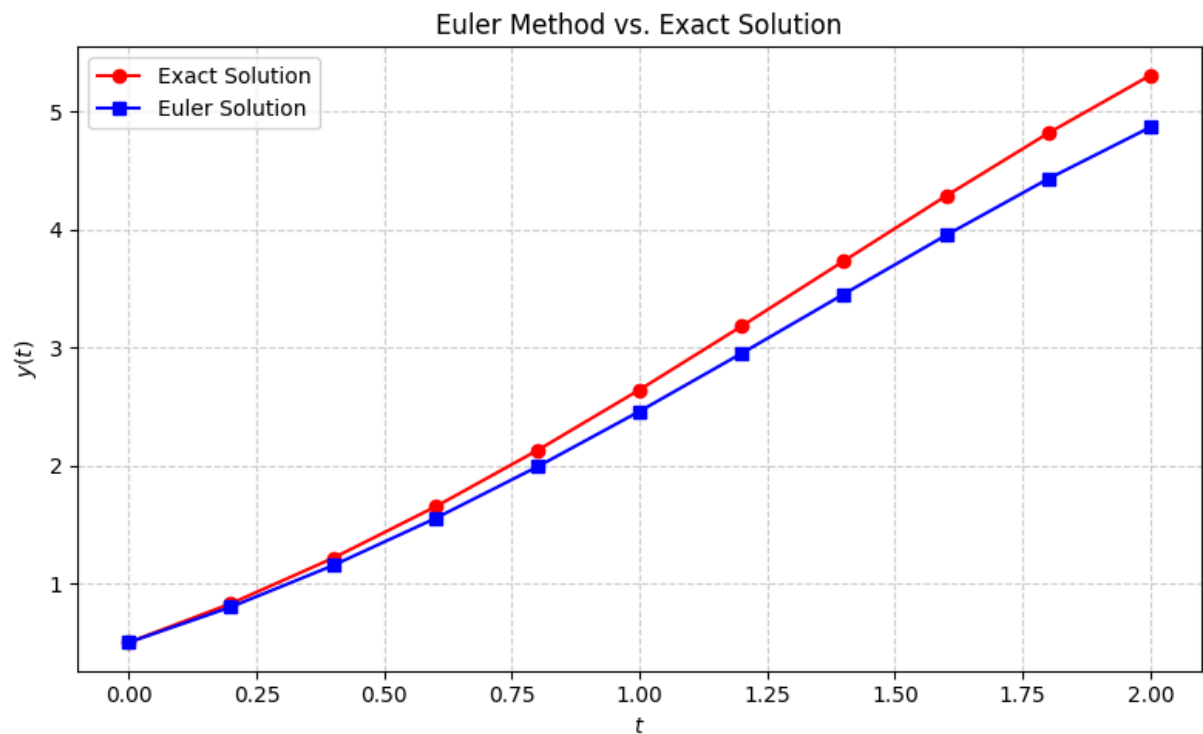
# Exact solution at grid points
y_exact = exact_y(t)
errors = np.abs(y_num - y_exact)

# Display results
print("-" * 54)
print(f"                Euler Method    h = {(b - a)/N:.1f}")
print("-" * 54)
print("t_i          y_i          y(t_i)      |y_i-y(t_i)|")
print("-" * 54)
for ti, yi, ye, err in zip(t, y_num, y_exact, errors):
    print(f"{ti:6.4f}    {yi:8.4f}    {ye:8.4f}    {err:8.4f}")

# Plot
plt.figure(figsize=(8, 5))
plt.plot(t, y_exact, 'ro-', label='Exact Solution', markersize=6)
plt.plot(t, y_num, 'bs-', label='Euler Solution', markersize=6)
plt.xlabel('$t$')
plt.ylabel('$y(t)$')
plt.title('Euler Method vs. Exact Solution')
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()

```

Euler Method $h = 0.2$			
t_i	y_i	$y(t_i)$	$ y_i - y(t_i) $
0.0000	0.5000	0.5000	0.0000
0.2000	0.8000	0.8293	0.0293
0.4000	1.1520	1.2141	0.0621
0.6000	1.5504	1.6489	0.0985
0.8000	1.9885	2.1272	0.1387
1.0000	2.4582	2.6409	0.1827
1.2000	2.9498	3.1799	0.2301
1.4000	3.4518	3.7324	0.2806
1.6000	3.9501	4.2835	0.3334
1.8000	4.4282	4.8152	0.3870
2.0000	4.8658	5.3055	0.4397



Plot of Euler's method solution and exact solution for different h values. The approximations become more accurate as the step size h gets smaller.

```
In [3]: import numpy as np
import matplotlib.pyplot as plt
# Define the ODE and exact solution
def f(t, y):
    return y - t**2 + 1

def exact_solution(t):
    return (t + 1)**2 - 0.5 * np.exp(t)

# Euler method (self-contained for this cell)
def euler_method(fun, a, b, y0, N):
    h = (b - a) / N
    t = np.linspace(a, b, N + 1)
```

```

y = np.empty(N + 1)
y[0] = y0
for i in range(N):
    y[i + 1] = y[i] + h * fun(t[i], y[i])
return t, y

# Parameters
a, b = 0.0, 2.0
y0 = 0.5
N_values = [10, 20, 50]

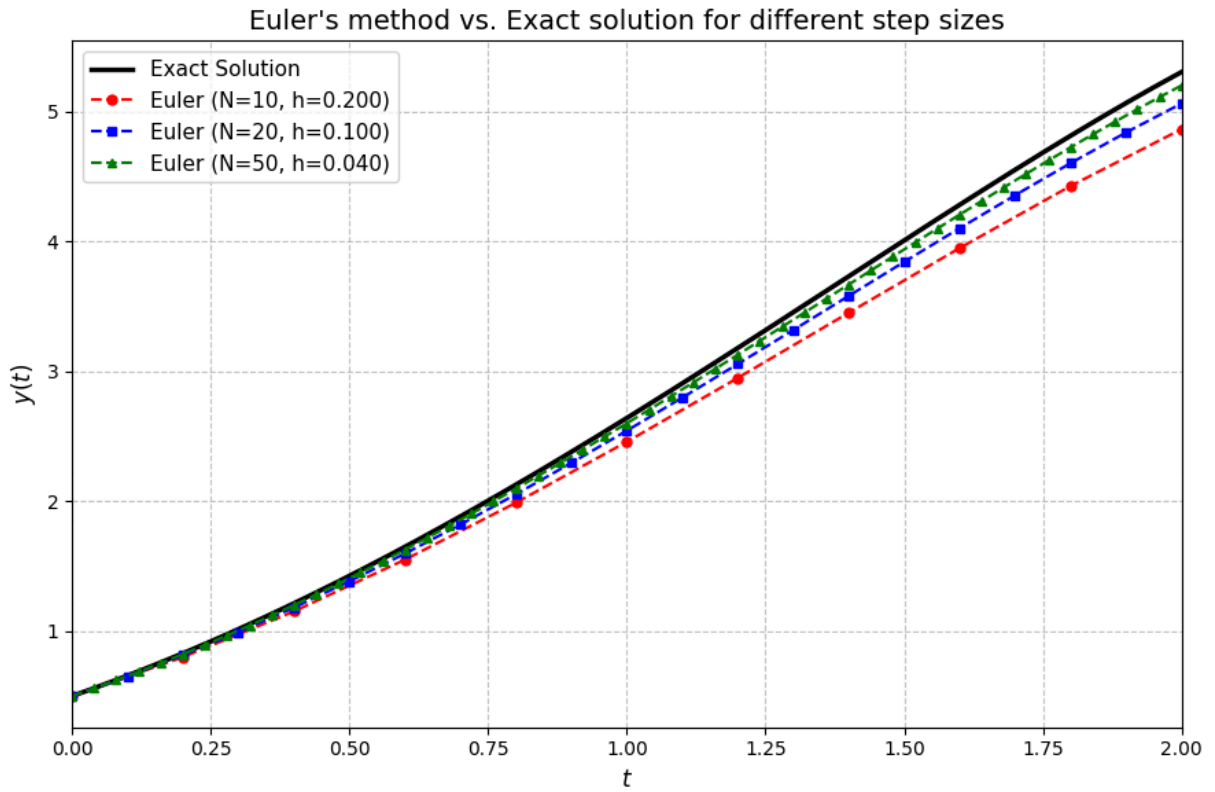
# Fine grid for smooth exact curve
t_fine = np.linspace(a, b, 400)
y_exact_fine = exact_solution(t_fine)

# Plot
plt.figure(figsize=(9, 6))
plt.plot(t_fine, y_exact_fine, 'k-', linewidth=2.5, label='Exact Solution')

colors = ['red', 'blue', 'green']
markers = ['o', 's', '^']

for N, color, marker in zip(N_values, colors, markers):
    t_num, y_num = euler_method(f, a, b, y0, N)
    h = (b - a) / N
    plt.plot(t_num, y_num,
             linestyle='--',
             marker=marker,
             color=color,
             markersize=5,
             label=f'Euler (N={N}, h={h:.3f})')
plt.xlabel('$t$', fontsize=12)
plt.ylabel('$y(t)$', fontsize=12)
plt.title("Euler's method vs. Exact solution for different step sizes", fontsize=14)
plt.legend(fontsize=11)
plt.grid(True, linestyle='--', alpha=0.7)
plt.xlim(a, b)
plt.tight_layout()
plt.show()

```



Error Bounds for Euler's Method

Although Euler's method is not accurate enough for practical use, it is sufficiently elementary to allow a clear analysis of the error it produces. The more accurate methods discussed later follow similar principles but involve greater complexity.

To derive an error bound for Euler's method, we first establish two computational lemmas.

Lemma

For all $x \geq -1$ and any positive integer m ,

$$0 \leq (1 + x)^m \leq e^{mx}.$$

Proof

By Taylor's Theorem applied to $f(x) = e^x$ with $x_0 = 0$ and $n = 1$,

$$e^x = 1 + x + \frac{1}{2}x^2e^\xi,$$

where ξ lies between 0 and x . Since $\frac{1}{2}x^2e^\xi \geq 0$, we have

$$1 + x \leq e^x.$$

Given $x \geq -1$, it follows that $1 + x \geq 0$. Raising both sides to the power m preserves the inequality:

$$(1+x)^m \leq (e^x)^m = e^{mx}.$$

The left inequality $(1+x)^m \geq 0$ holds because $1+x \geq 0$. ■

Lemma

Let $s > 0$, $t > 0$, and let $\{a_i\}_{i=0}^k$ be a sequence such that

$$a_0 \geq -\frac{t}{s}, \quad \text{and} \quad a_{i+1} \leq (1+s)a_i + t, \quad \text{for each } i = 0, 1, \dots, k-1.$$

Then

$$a_{i+1} \leq e^{(i+1)s} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s}, \quad \text{for } i = 0, 1, \dots, k-1.$$

Proof

We proceed by unwinding the recurrence:

$$\begin{aligned} a_{i+1} &\leq (1+s)a_i + t \\ &\leq (1+s)[(1+s)a_{i-1} + t] + t = (1+s)^2 a_{i-1} + [1 + (1+s)]t \\ &\leq (1+s)^3 a_{i-2} + [1 + (1+s) + (1+s)^2]t \\ &\vdots \\ &\leq (1+s)^{i+1} a_0 + \left(\sum_{j=0}^i (1+s)^j \right) t. \end{aligned}$$

The sum is a geometric series:

$$\sum_{j=0}^i (1+s)^j = \frac{(1+s)^{i+1} - 1}{s}.$$

Thus,

$$a_{i+1} \leq (1+s)^{i+1} a_0 + \frac{(1+s)^{i+1} - 1}{s} t = (1+s)^{i+1} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s}.$$

By the previous lemma, $(1+s)^{i+1} \leq e^{(i+1)s}$ for $s > -1$ (which holds since $s > 0$).

Therefore,

$$a_{i+1} \leq e^{(i+1)s} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s}.$$

■

Theorem (Global Error Bound for Euler's Method)

Suppose f is continuous and satisfies a Lipschitz condition in y with constant L on the strip

$$D = \{(t, y) \mid a \leq t \leq b, -\infty < y < \infty\}.$$

Let $y(t)$ be the unique solution to the IVP

$$y' = f(t, y), \quad y(a) = \alpha, \quad a \leq t \leq b,$$

and assume $y \in C^2[a, b]$. Suppose there exists a constant M such that

$$|y''(t)| \leq M, \quad \text{for all } t \in [a, b].$$

Let $w_0, w_1, \dots, w_N$ be the approximations generated by Euler's method with step size $h = \frac{b-a}{N}$. Then, for each $i = 0, 1, \dots, N$,

$$|y(t_i) - w_i| \leq \frac{hM}{2L} \left(e^{L(t_i-a)} - 1 \right).$$

Proof

For $i = 0$, $t_0 = a$ and $w_0 = \alpha = y(a)$, so $|y(t_0) - w_0| = 0$, and the bound holds.

From Taylor's Theorem, for each $i = 0, 1, \dots, N-1$, there exists $\xi_i \in (t_i, t_{i+1})$ such that

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i).$$

Euler's method gives

$$w_{i+1} = w_i + hf(t_i, w_i).$$

Define the global error $e_i = y(t_i) - w_i$. Subtracting:

$$e_{i+1} = e_i + h[f(t_i, y(t_i)) - f(t_i, w_i)] + \frac{h^2}{2} y''(\xi_i).$$

Taking absolute values and using the Lipschitz condition $|f(t, y(t_i)) - f(t, w_i)| \leq L|e_i|$ and $|y''(\xi_i)| \leq M$:

$$|e_{i+1}| \leq |e_i| + hL|e_i| + \frac{h^2M}{2} = (1 + hL)|e_i| + \frac{h^2M}{2}.$$

This recurrence matches the form in the previous lemma with

$$s = hL, \quad t = \frac{h^2M}{2}, \quad a_i = |e_i|.$$

Since $a_0 = |e_0| = 0 \geq -\frac{t}{s} = -\frac{hM}{2L}$, the lemma applies. Hence,

$$|e_{i+1}| \leq e^{(i+1)hL} \left(0 + \frac{h^2 M/2}{hL} \right) - \frac{h^2 M/2}{hL} = \frac{hM}{2L} (e^{(i+1)hL} - 1).$$

Because $t_{i+1} = a + (i+1)h$, we have $(i+1)h = t_{i+1} - a$, so

$$|y(t_{i+1}) - w_{i+1}| \leq \frac{hM}{2L} (e^{L(t_{i+1}-a)} - 1).$$

Reindexing yields the result for all $i = 0, 1, \dots, N$. ■

The weakness of the global error bound theorem lies in the requirement that a constant M satisfying $|y''(t)| \leq M$ on $[a, b]$ be known. In practice, this is often unavailable—since the exact solution $y(t)$ is precisely what we seek to approximate.

However, if $f \in C^1(D)$, then by the chain rule:

$$y''(t) = \frac{d}{dt} f(t, y(t)) = \frac{\partial f}{\partial t}(t, y(t)) + \frac{\partial f}{\partial y}(t, y(t)) \cdot f(t, y(t)).$$

Thus, a bound on $|y''(t)|$ can sometimes be obtained *without* knowing $y(t)$ explicitly—by bounding f , $f_t = \partial f / \partial t$, and $f_y = \partial f / \partial y$ over a suitable region (e.g., a rectangle containing the solution trajectory).

For instance, if $|f| \leq K$, $|f_t| \leq K_1$, and $|f_y| \leq K_2$ on D , then

$$|y''(t)| \leq K_1 + K_2 K,$$

so we may take $M = K_1 + K_2 K$.

This analysis confirms that Euler's method is **first-order accurate**: the global error satisfies

$$\max_{0 \leq i \leq N} |y(t_i) - w_i| = O(h) \quad \text{as } h \rightarrow 0.$$

Because the error decreases only linearly with step size, very small h is needed for high accuracy—leading to increased computational cost and potential round-off accumulation. This motivates the development of **higher-order methods**, such as:

- Taylor series methods (using higher derivatives),
- Runge–Kutta methods (e.g., classical RK4, order 4),
- Multistep methods (e.g., Adams–Bashforth, Adams–Moulton),

which achieve superior accuracy *without* requiring explicit evaluation of higher derivatives of f .

Example

The solution to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

was approximated in a previous example using Euler's method with $h = 0.2$. Use the error bound from the Global Error Theorem to estimate the approximation errors and compare them to the actual errors.

Solution

Here, $f(t, y) = y - t^2 + 1$, so

$$\frac{\partial f}{\partial y} = 1 \quad \Rightarrow \quad L = 1.$$

The exact solution is

$$y(t) = (t + 1)^2 - 0.5e^t,$$

hence

$$y'(t) = 2(t + 1) - 0.5e^t, \quad y''(t) = 2 - 0.5e^t.$$

On $[0, 2]$, $y''(t)$ is decreasing, so its maximum absolute value occurs at $t = 2$:

$$|y''(t)| \leq M = \max_{t \in [0, 2]} |2 - 0.5e^t| = 0.5e^2 - 2 \approx 1.6945.$$

The error bound from the theorem is:

$$|y(t_i) - w_i| \leq \frac{hM}{2L} (e^{L(t_i-a)} - 1) = 0.1 (0.5e^2 - 2) (e^{t_i} - 1).$$

For example:

$$\begin{aligned} |y(0.2) - w_1| &\leq 0.1 (0.5e^2 - 2) (e^{0.2} - 1) \approx 0.03752, \\ |y(0.4) - w_2| &\leq 0.1 (0.5e^2 - 2) (e^{0.4} - 1) \approx 0.08334. \end{aligned}$$

The table below compares the **actual errors** (computed from the exact solution) with the **theoretical error bounds**:

```
In [4]: import numpy as np
# Define ODE and exact solution
def f(t, y):
    return y - t**2 + 1
def y_exact(t):
    return (t + 1)**2 - 0.5 * np.exp(t)
```

```

# Parameters
a, b = 0.0, 2.0
h = 0.2
N = int(round((b - a) / h))
t = np.linspace(a, b, N + 1)

# Euler method
def euler_method(fun, t_grid, y0):
    w = np.empty_like(t_grid)
    w[0] = y0
    for i in range(len(t_grid) - 1):
        h_i = t_grid[i+1] - t_grid[i]
        w[i + 1] = w[i] + h_i * fun(t_grid[i], w[i])
    return w

# Compute solution and errors
w = euler_method(f, t, y0=0.5)
y_true = y_exact(t)
actual_error = np.abs(y_true - w)

# Error bound
L = 1.0
M = 0.5 * np.exp(2) - 2
error_bound = (h * M / (2 * L)) * (np.exp(L * (t - a)) - 1)

# Print table
print(f"{'t_i':>5} {'Actual Error':>12} {'Error Bound':>12}")
print("-" * 32)
for ti, err, bnd in zip(t, actual_error, error_bound):
    print(f"{'ti':5.1f} {'err':12.5f} {'bnd':12.5f}")

```

t_i	Actual Error	Error Bound
0.0	0.00000	0.00000
0.2	0.02930	0.03752
0.4	0.06209	0.08334
0.6	0.09854	0.13931
0.8	0.13875	0.20767
1.0	0.18268	0.29117
1.2	0.23013	0.39315
1.4	0.28063	0.51771
1.6	0.33336	0.66985
1.8	0.38702	0.85568
2.0	0.43969	1.08264

Observation: The theoretical bound is always larger than the actual error — and increasingly conservative as t grows. This is typical: the bound guarantees *worst-case* behavior, while the actual solution may be smoother or better behaved.

The key takeaway is that the bound is **linear in h** . Hence, halving h roughly halves the bound — confirming that Euler’s method is **first-order accurate**.

Lab Exercise: Euler's Method Practice

Use Euler's method to approximate the solutions for each of the following initial-value problems. Modify the earlier Euler implementation for each case.

a. $y' = \frac{2-2ty}{t^2+1}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$. Exact solution: $y(t) = \frac{2t+1}{t^2+1}$

b. $y' = \frac{y^2}{1+t}$, $1 \leq t \leq 2$, $y(1) = -\frac{1}{\ln 2}$, $h = 0.1$. Exact solution: $y(t) = -\frac{1}{\ln(1+t)}$

c. $y' = \frac{y^2+y}{t}$, $1 \leq t \leq 3$, $y(1) = -2$, $h = 0.2$. Exact solution: $y(t) = \frac{2t}{1-2t}$

d. $y' = -ty + \frac{4t}{y}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$. Exact solution:
 $y(t) = \sqrt{4 - 3e^{-t^2}}$

✦ **Task:** For each problem,

1. Implement Euler's method,
2. Compute the numerical solution,
3. Evaluate the exact solution at the grid points,
4. Compute and report the absolute error at each step (or at the final point),
5. Plot numerical vs. exact solution.

Taylor's Methods

Consider the initial-value problem (IVP)

$$\begin{cases} y'(t) = f(t, y(t)), & a \leq t \leq b, \\ y(a) = y_0, \end{cases}$$

where $f : [a, b] \times \mathbb{R} \rightarrow \mathbb{R}$ is assumed sufficiently smooth (e.g., $f \in C^{n+1}$).

Let $y(t)$ be the exact solution. By Taylor's theorem with Lagrange remainder, for any $t_i \in [a, b]$ and step size $h > 0$ such that $t_{i+1} = t_i + h \in [a, b]$, we have

$$y(t_{i+1}) = \sum_{k=0}^n \frac{h^k}{k!} y^{(k)}(t_i) + R_{n+1}(t_i),$$

where the remainder term is

$$R_{n+1}(t_i) = \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_i), \quad \text{for some } \xi_i \in (t_i, t_{i+1}).$$

To construct a numerical method, we express the derivatives $y^{(k)}(t_i)$ in terms of f and its partial derivatives.

From the IVP, the first derivative is:

$$y'(t) = f(t, y(t)).$$

Differentiating with respect to t and applying the chain rule gives the second derivative:

$$\begin{aligned} y''(t) &= \frac{d}{dt} f(t, y(t)) = \frac{\partial f}{\partial t}(t, y(t)) + \frac{\partial f}{\partial y}(t, y(t)) \cdot y'(t) \\ &= f_t(t, y(t)) + f_y(t, y(t)) f(t, y(t)). \end{aligned}$$

This motivates the definition of the **total derivative operator** (also called the *Lie derivative* along the flow):

Definition (Total Derivative Operator)

Define the operator $\mathcal{D}$ acting on sufficiently smooth functions $g(t, y)$ by

$$\mathcal{D}g := \frac{\partial g}{\partial t} + y' \frac{\partial g}{\partial y}.$$

Using $\mathcal{D}$, higher-order time derivatives of the solution can be expressed compactly:

Theorem

For $k \geq 1$, the k -th derivative of the solution satisfies

$$y^{(k)}(t) = \mathcal{D}^{k-1} f(t, y(t)).$$

Proof

By induction. The case $k = 1$ holds since $y' = f = \mathcal{D}^0 f$.

Assume the formula holds for some k . Then

$$y^{(k+1)}(t) = \frac{d}{dt} [\mathcal{D}^{k-1} f(t, y(t))] = \mathcal{D}(\mathcal{D}^{k-1} f)(t, y(t)) = \mathcal{D}^k f(t, y(t)),$$

as required. ■

We define successive total derivatives of f :

Definition

Let $f^{[k]}(t, y)$ denote the k -th total derivative of f :

$$\begin{aligned} f^{[0]}(t, y) &:= f(t, y), \\ f^{[k]}(t, y) &:= \mathcal{D}^k f(t, y), \quad k = 1, 2, \dots \end{aligned}$$

Then $y^{(k+1)}(t) = f^{[k]}(t, y(t))$.

Substituting into the Taylor expansion gives

$$y(t_{i+1}) = y(t_i) + \sum_{k=1}^n \frac{h^k}{k!} f^{[k-1]}(t_i, y(t_i)) + \frac{h^{n+1}}{(n+1)!} f^{[n]}(\xi_i, y(\xi_i)).$$

A numerical method is obtained by:

- Approximating $y(t_i)$ by a computable value y_i ,
- Dropping the remainder term.

Definition (Taylor's Method of Order n)

Given $t_0 = a$, y_0 , and step size $h = (b - a)/N$, compute for $i = 0, 1, \dots, N - 1$:

$$y_{i+1} = y_i + \sum_{k=1}^n \frac{h^k}{k!} f^{[k-1]}(t_i, y_i), \quad \text{with } t_{i+1} = t_i + h.$$

Explicitly, the first few orders are:

- **Order 1 (Euler's method):**

$$y_{i+1} = y_i + h f(t_i, y_i).$$

- **Order 2:**

$$y_{i+1} = y_i + h f(t_i, y_i) + \frac{h^2}{2} \left[f_t(t_i, y_i) + f_y(t_i, y_i) f(t_i, y_i) \right].$$

- **Order 3:**

$$y_{i+1} = y_i + h f + \frac{h^2}{2} (f_t + f_y f) + \frac{h^3}{6} \left[f_{tt} + 2f_{ty} f + f_{yy} f^2 + f_y f_t + f_y^2 f \right],$$

- **Order 4:**

$$\begin{aligned} y_{i+1} = y_i + h f + \frac{h^2}{2} (f_t + f_y f) + \frac{h^3}{6} (f_{tt} + 2f_{ty} f + f_{yy} f^2 + f_y f_t + f_y^2 f) \\ + \frac{h^4}{24} \left[f_{ttt} + 3f_{tty} f + 3f_{tyy} f^2 + f_{yyy} f^3 + 3f_{tt} f_y + 3f_{ty} (2f_t + 3f_y f) + f_{yy} (3f_t + f_t(f_{ty} + f_{yy} f + f_y^2) + f_y^2(f_t + f_y f) + f_y(f_{tt} + 2f_{ty} f + f_{yy} f^2)) \right], \end{aligned}$$

where all partial derivatives of f are evaluated at (t_i, y_i) .



Theorem (Local Truncation Error)

The local truncation error (LTE) of Taylor's method of order n satisfies

$$\text{LTE}_{i+1} := y(t_{i+1}) - \left(y(t_i) + \sum_{k=1}^n \frac{h^k}{k!} f^{[k-1]}(t_i, y(t_i)) \right) = \frac{h^{n+1}}{(n+1)!} f^{[n]}(\xi_i, y(\xi_i)) = \mathcal{O}(h^{n+1})$$

provided $f^{[n]}$ is bounded on a neighborhood of the solution.

Theorem (Global Truncation Error)

Assume f is Lipschitz continuous in y and the total derivatives $f^{[k]}$ are bounded for $k = 0, \dots, n - 1$. Then the global error satisfies

$$\max_{0 \leq i \leq N} |y(t_i) - y_i| = \mathcal{O}(h^n),$$

i.e., Taylor's method of order n is convergent of order n .

Remark

Euler's method corresponds to $n = 1$:

- $\text{LTE} = \mathcal{O}(h^2)$,
- $\text{GTE} = \mathcal{O}(h)$.

Example

Apply Taylor's methods of order 1, 2, and 4 to solve the IVP:

$$y' = f(t, y) = y - t^2 + 1, \quad y(0) = \frac{1}{2}, \quad 0 \leq t \leq 2.$$

Solution

Let $f(t, y) = y - t^2 + 1$. Using the total derivative operator $\mathcal{D} = \partial_t + f \partial_y$, compute successive derivatives:

$$\begin{aligned} y' &= f = y - t^2 + 1, \\ y'' &= \mathcal{D}f = (-2t) + (1)(y - t^2 + 1) = y - t^2 - 2t + 1, \\ y''' &= \mathcal{D}(y'') = (-2t - 2) + (1)(y - t^2 - 2t + 1) = y - t^2 - 4t - 1, \\ y^{(4)} &= \mathcal{D}(y''') = (-2t - 4) + (1)(y - t^2 - 4t - 1) = y - t^2 - 6t - 5. \end{aligned}$$

Let (t_n, y_n) be the current point and h the step size. Then:

- **Order 1 (Euler):**

$$y_{n+1} = y_n + h(y_n - t_n^2 + 1).$$

- **Order 2:**

$$\begin{aligned} y_{n+1} &= y_n + h y'_n + \frac{h^2}{2} y''_n \\ &= y_n + h(y_n - t_n^2 + 1) + \frac{h^2}{2}(y_n - t_n^2 - 2t_n + 1). \end{aligned}$$

- **Order 4:**

$$y_{n+1} = y_n + h(y_n - t_n^2 + 1) + \frac{h^2}{2}(y_n - t_n^2 - 2t_n + 1) + \frac{h^3}{6}(y_n - t_n^2 - 4t_n - 1) + \frac{h^4}{24}(y_n - t_n^2 - 6t_n - 5).$$

The exact solution is

$$y(t) = (t + 1)^2 - \frac{1}{2}e^t,$$

which can be used to compute the error $e_n = |y(t_n) - y_n|$.

```
In [5]: import numpy as np
import matplotlib.pyplot as plt
# ----- Taylor Methods Implementations
def taylor2(fun, Dfun, a, b, y0, N):
    """
    Taylor's method of order 2.
    fun : f(t, y)
    Dfun : Df = f_t + f_y * f (i.e., y'')
    """
    h = (b - a) / N
    t = np.linspace(a, b, N + 1)
    y = np.empty(N + 1)
    y[0] = y0
    for i in range(N):
        y[i + 1] = (y[i]
                    + h * fun(t[i], y[i])
                    + (h**2 / 2) * Dfun(t[i], y[i]))
    return t, y

def taylor4(fun, Dfun, DDfun, DDDfun, a, b, y0, N):
    """
    Taylor's method of order 4.
    fun      : f          = y'
    Dfun      : Df         = y''
    DDfun     : D^2 f      = y'''
    DDDfun    : D^3 f      = y^(4)
    """
    h = (b - a) / N
    t = np.linspace(a, b, N + 1)
    y = np.empty(N + 1)
    y[0] = y0
    for i in range(N):
        y[i + 1] = (y[i]
                    + h * fun(t[i], y[i])
                    + (h**2 / 2) * Dfun(t[i], y[i])
                    + (h**3 / 6) * DDfun(t[i], y[i])
                    + (h**4 / 24) * DDDfun(t[i], y[i]))
    return t, y

# Euler method (for comparison)
def euler_method(fun, a, b, y0, N):
    h = (b - a) / N
    t = np.linspace(a, b, N + 1)
    y = np.empty(N + 1)
    y[0] = y0
    for i in range(N):
```

```

        y[i + 1] = y[i] + h * fun(t[i], y[i])
    return t, y
# ----- Problem Setup -----
fun = lambda t, y: y - t**2 + 1
Dfun = lambda t, y: y - t**2 - 2*t + 1          # y''
DDfun = lambda t, y: y - t**2 - 4*t - 1         # y'''
DDDfun = lambda t, y: y - t**2 - 6*t - 5        # y^(4)

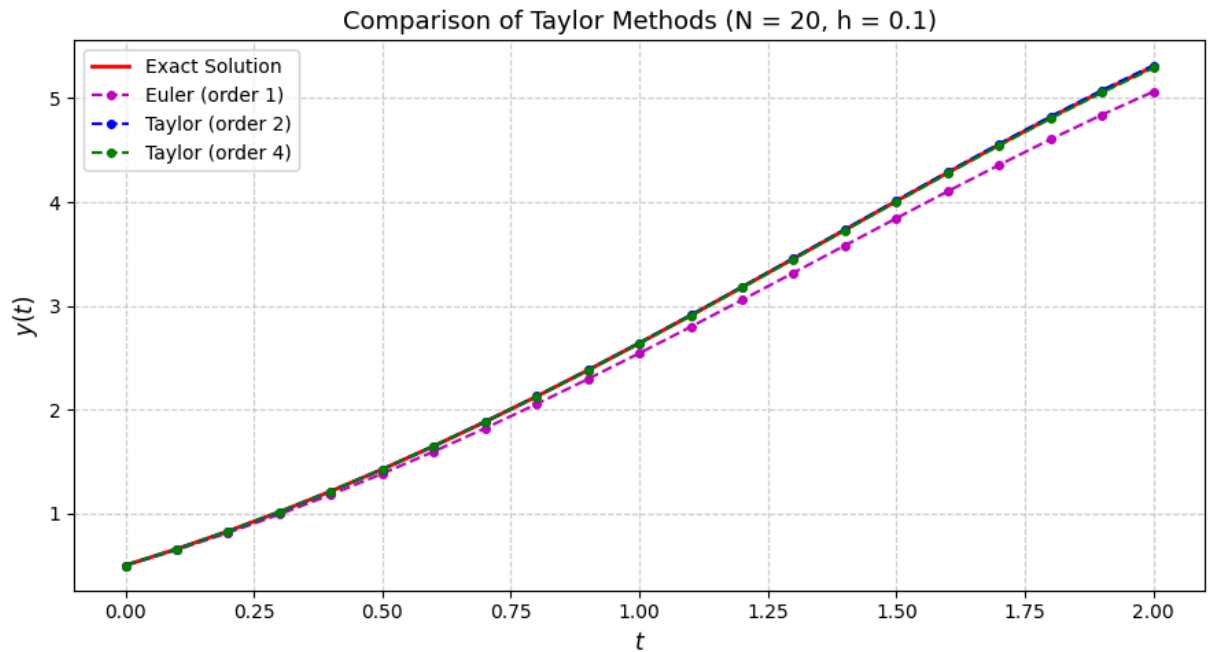
a, b = 0.0, 2.0
y0 = 0.5
N = 20
# Solve
t_eul, y_eul = euler_method(fun, a, b, y0, N)
t_t2, y_t2 = taylor2(fun, Dfun, a, b, y0, N)
t_t4, y_t4 = taylor4(fun, Dfun, DDfun, DDDfun, a, b, y0, N)
# Exact solution
exact = lambda t: (t + 1)**2 - 0.5 * np.exp(t)
y_exact = exact(t_eul) # all t-vectors are identical (uniform grid)
# Errors
err_eul = np.abs(y_eul - y_exact)
err_t2 = np.abs(y_t2 - y_exact)
err_t4 = np.abs(y_t4 - y_exact)

# ----- Display Table -----
print("-" * 92)
print(f"{'':<6} {'Exact':<10} {'Euler':<15} {'Taylor(2)':<15} {'Taylor(4)':<15}")
print(f"{'t_i':<6} {'y(t_i)':<10} {'y_i':<8} {'|e_i|':<10} {'y_i':<8} {'|e_i|':<10}")
print("-" * 92)
for i in range(len(t_eul)):
    print(f"{t_eul[i]:<6.1f} "
          f"{y_exact[i]:<10.5f} "
          f"{y_eul[i]:<8.5f} {err_eul[i]:<10.2e} "
          f"{y_t2[i]:<8.5f} {err_t2[i]:<10.2e} "
          f"{y_t4[i]:<8.5f} {err_t4[i]:<10.2e}")

# ----- Plot -----
plt.figure(figsize=(9, 5))
plt.plot(t_eul, y_exact, 'r-', linewidth=2, label='Exact Solution')
plt.plot(t_eul, y_eul, 'mo--', markersize=4, label='Euler (order 1)')
plt.plot(t_t2, y_t2, 'bo--', markersize=4, label='Taylor (order 2)')
plt.plot(t_t4, y_t4, 'go--', markersize=4, label='Taylor (order 4)')
plt.xlabel('$t$', fontsize=12)
plt.ylabel('$y(t)$', fontsize=12)
plt.title('Comparison of Taylor Methods (N = 20, h = 0.1)', fontsize=13)
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()

```

t_i	Exact y(t_i)	Euler y_i	e_i	Taylor(2) y_i	e_i	Taylor(4) y_i	e_i
0.0	0.50000	0.50000	0.00e+00	0.50000	0.00e+00	0.50000	0.00e+00
0.1	0.65741	0.65000	7.41e-03	0.65750	8.55e-05	0.65740	1.66e-05
0.2	0.82930	0.81400	1.53e-02	0.82949	1.89e-04	0.82923	7.00e-05
0.3	1.01507	0.99140	2.37e-02	1.01538	3.13e-04	1.01491	1.64e-04
0.4	1.21409	1.18154	3.25e-02	1.21455	4.61e-04	1.21378	3.03e-04
0.5	1.42564	1.38369	4.19e-02	1.42628	6.37e-04	1.42515	4.91e-04
0.6	1.64894	1.59706	5.19e-02	1.64979	8.45e-04	1.64821	7.35e-04
0.7	1.88312	1.82077	6.24e-02	1.88421	1.09e-03	1.88209	1.04e-03
0.8	2.12723	2.05385	7.34e-02	2.12861	1.38e-03	2.12582	1.41e-03
0.9	2.38020	2.29523	8.50e-02	2.38191	1.71e-03	2.37834	1.85e-03
1.0	2.64086	2.54375	9.71e-02	2.64296	2.10e-03	2.63848	2.38e-03
1.1	2.90792	2.79813	1.10e-01	2.91047	2.55e-03	2.90492	3.00e-03
1.2	3.17994	3.05694	1.23e-01	3.18302	3.08e-03	3.17623	3.71e-03
1.3	3.45535	3.31864	1.37e-01	3.45904	3.69e-03	3.45081	4.54e-03
1.4	3.73240	3.58150	1.51e-01	3.73679	4.39e-03	3.72691	5.49e-03
1.5	4.00916	3.84365	1.66e-01	4.01435	5.19e-03	4.00258	6.57e-03
1.6	4.28348	4.10302	1.80e-01	4.28960	6.12e-03	4.27568	7.81e-03
1.7	4.55303	4.35732	1.96e-01	4.56021	7.19e-03	4.54382	9.20e-03
1.8	4.81518	4.60405	2.11e-01	4.82359	8.41e-03	4.80439	1.08e-02
1.9	5.06705	4.84045	2.27e-01	5.07686	9.81e-03	5.05449	1.26e-02
2.0	5.30547	5.06350	2.42e-01	5.31688	1.14e-02	5.29090	1.46e-02



Reduce N to 5 or 10 and observe how Taylor-4 remains accurate while Euler deteriorates.

Remark

For this linear problem, higher total derivatives stabilize: $f^{[k]} = f^{[2]}$ for all $k \geq 2$, enabling highly accurate high-order Taylor approximations even with moderate h .

Remark

A major drawback of Taylor series methods is that they require the explicit computation of higher-order derivatives of $f(t, y)$, i.e., expressions such as

$$\frac{d^k}{dt^k} f(t, y(t)) = \mathcal{D}^k f(t, y), \quad k = 1, 2, \dots, p-1,$$

where $\mathcal{D} = \partial_t + f(t, y) \partial_y$ is the total derivative operator. These derivatives become increasingly complex and cumbersome to derive and implement even for modest orders p and are often infeasible for nonlinear or high-dimensional systems.

Lab Work: Taylor's Method of Order 2, 3, and 4

Implement Taylor's method of order 2, 3, and 4 to approximate the solutions of the following initial value problems (IVPs).

For each case, compare the numerical approximations with the exact solution. Use the step sizes indicated.

- a. $y' = \frac{2-2ty}{t^2+1}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$. Exact solution: $y(t) = \frac{2t+1}{t^2+1}$
- b. $y' = \frac{y^2}{1+t}$, $1 \leq t \leq 2$, $y(1) = -\frac{1}{\ln 2}$, $h = 0.1$. Exact solution: $y(t) = -\frac{1}{\ln(1+t)}$
- c. $y' = \frac{y^2+y}{t}$, $1 \leq t \leq 3$, $y(1) = -2$, $h = 0.2$. Exact solution: $y(t) = \frac{2t}{1-2t}$
- d. $y' = -ty + \frac{4t}{y}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$. Exact solution: $y(t) = \sqrt{4 - 3e^{-t^2}}$

✦ **Task:** For each problem,

1. Implement Taylor's methods,
2. Compute the numerical solution,
3. Evaluate the exact solution at the grid points,
4. Compute and report the absolute error at each step (or at the final point),
5. Plot numerical vs. exact solution.

Runge–Kutta Methods

The Taylor methods introduced previously possess the desirable property of high-order local truncation error.

However, their practical applicability is limited by a significant drawback: they require the explicit computation and evaluation of higher-order partial derivatives of $f(t, y)$ (e.g., f_t , f_y , f_{tt} , f_{ty} , etc.).

For most nonlinear or moderately complex problems, deriving and implementing these derivatives is:

- algebraically cumbersome,
- error-prone, and
- computationally inefficient.

Consequently, Taylor methods are rarely used in practice beyond low-order implementations.

To overcome this limitation, we now introduce the **Runge–Kutta methods**.

These methods attain the same high order of local truncation error as their Taylor counterparts, yet **avoid the need to compute analytical derivatives of f** .

Instead, they achieve high-order accuracy by evaluating f multiple times per step at **strategically chosen intermediate points**.

Theoretical Foundation

The justification for Runge–Kutta methods relies on multivariate Taylor expansions. The following theorem formalizes the two-variable Taylor polynomial and its remainder.

Theorem

Suppose that $f(t, y)$ and all its partial derivatives of order $\leq n + 1$ are continuous on

$$D = \{(t, y) \mid a \leq t \leq b, c \leq y \leq d\},$$

and let $(t_0, y_0) \in D$.

Then for every $(t, y) \in D$, there exist ξ between t and t_0 , and μ between y and y_0 , such that

$$f(t, y) = P_n(t, y) + R_n(t, y),$$

where $P_n(t, y)$ is the n th-order Taylor polynomial of f about (t_0, y_0) , and $R_n(t, y)$ is the remainder.

The n th Taylor Polynomial in Two Variables

$$\begin{aligned} P_n(t, y) = & f(t_0, y_0) + (t - t_0) \frac{\partial f}{\partial t}(t_0, y_0) + (y - y_0) \frac{\partial f}{\partial y}(t_0, y_0) \\ & + \frac{(t - t_0)^2}{2} \frac{\partial^2 f}{\partial t^2}(t_0, y_0) + (t - t_0)(y - y_0) \frac{\partial^2 f}{\partial t \partial y}(t_0, y_0) + \frac{(y - y_0)^2}{2} \frac{\partial^2 f}{\partial y^2}(t_0, y_0) \\ & + \cdots + \frac{1}{n!} \sum_{j=0}^n \binom{n}{j} (t - t_0)^{n-j} (y - y_0)^j \frac{\partial^n f}{\partial t^{n-j} \partial y^j}(t_0, y_0). \end{aligned}$$



Remainder Term

$$R_n(t, y) = \frac{1}{(n+1)!} \sum_{j=0}^{n+1} \binom{n+1}{j} (t - t_0)^{n+1-j} (y - y_0)^j \frac{\partial^{n+1} f}{\partial t^{n+1-j} \partial y^j}(\xi, \mu).$$

Here, (ξ, μ) lies in the rectangle with corners at (t_0, y_0) and (t, y) .

✓ **Key insight for Runge–Kutta:** By matching the Taylor expansion of the *numerical solution* with P_n **up to order** n , while using only evaluations of f (not its derivatives), we can construct derivative-free high-order methods.

Runge–Kutta Methods of Order Two

The goal is to construct a derivative-free approximation to the second-order Taylor term:

$$T^{(2)}(t, y) = f(t, y) + \frac{h}{2} f'(t, y),$$

where

$$f'(t, y) = \frac{\partial f}{\partial t}(t, y) + \frac{\partial f}{\partial y}(t, y) \cdot f(t, y).$$

So,

$$T^{(2)}(t, y) = f(t, y) + \frac{h}{2} \frac{\partial f}{\partial t}(t, y) + \frac{h}{2} f(t, y) \frac{\partial f}{\partial y}(t, y).$$

We seek constants a_1, α_1, β_1 such that

$$a_1 f(t + \alpha_1, y + \beta_1)$$

matches $T^{(2)}(t, y)$ up to $O(h^2)$.

Using the first-order 2D Taylor expansion:

$$\begin{aligned} a_1 f(t + \alpha_1, y + \beta_1) &= a_1 f(t, y) + a_1 \alpha_1 \frac{\partial f}{\partial t}(t, y) + a_1 \beta_1 \frac{\partial f}{\partial y}(t, y) \\ &\quad + a_1 \cdot R_1(t + \alpha_1, y + \beta_1), \end{aligned}$$

where the remainder is:

$$R_1 = \frac{\alpha_1^2}{2} f_{tt}(\xi, \mu) + \alpha_1 \beta_1 f_{ty}(\xi, \mu) + \frac{\beta_1^2}{2} f_{yy}(\xi, \mu),$$

for some (ξ, μ) near (t, y) .

Matching coefficients with $T^{(2)}$ gives:

$$\begin{cases} a_1 = 1, \\ a_1 \alpha_1 = \frac{h}{2}, \\ a_1 \beta_1 = \frac{h}{2} f(t, y). \end{cases} \Rightarrow a_1 = 1, \alpha_1 = \frac{h}{2}, \beta_1 = \frac{h}{2} f(t, y).$$

Thus,

$$T^{(2)}(t, y) = f\left(t + \frac{h}{2}, y + \frac{h}{2} f(t, y)\right) - R_1,$$

and since $\alpha_1, \beta_1 = O(h)$, the remainder satisfies

$$R_1 = O(h^2),$$

provided the second partial derivatives of f are bounded.

✓ **Conclusion:** This gives a *second-order* method **without computing any derivatives of f** .

◆ Midpoint Method (RK2)

Replacing $T^{(2)}$ in the Taylor step $w_{i+1} = w_i + h T^{(2)}(t_i, w_i)$ yields:

$$\begin{aligned} w_0 &= \alpha, \\ w_{i+1} &= w_i + h f\left(t_i + \frac{h}{2}, w_i + \frac{h}{2} f(t_i, w_i)\right), \quad i = 0, 1, \dots, N-1. \end{aligned}$$

This is known as the **Midpoint Method**, a classical second-order Runge–Kutta scheme.

Why Not Higher Order with Just One Stage?

The expression $a_1 f(t + \alpha_1, y + \beta_1)$ has only **three free parameters**, all used up to match $T^{(2)}$.

To reach third order, more flexibility is needed.

A natural extension is a **two-stage** form:

$$a_1 f(t, y) + a_2 f(t + \alpha_2, y + \delta_2 f(t, y)),$$

which introduces **four parameters**: $a_1, a_2, \alpha_2, \delta_2$.

However, even this form cannot match *all* terms in $T^{(3)}$ — particularly, it fails to replicate mixed terms like

$$\frac{h^2}{6} \cdot 2 f_y(t, y) f(t, y),$$

which arise from $f''(t, y)$.

→ So, all such 2-stage methods are limited to **local truncation error** $O(h^2)$ — i.e., *second-order global accuracy*.

But the extra degree of freedom allows **multiple distinct second-order methods**.

◆ Modified Euler Method (Heun's Method)

One important choice is:

$$a_1 = a_2 = \frac{1}{2}, \quad \alpha_2 = \delta_2 = h.$$

This leads to the **Modified Euler Method** (also called **Heun's method**):

$$w_0 = \alpha, \\ w_{i+1} = w_i + \frac{h}{2} \left[f(t_i, w_i) + f\left(t_i + h, w_i + h f(t_i, w_i)\right) \right], \quad i = 0, 1, \dots, N-1.$$

↔ Interpretation:

- Predict $\tilde{w}_{i+1} = w_i + h f(t_i, w_i)$ (Euler forward),
- Then correct using the average slope:

$$w_{i+1} = w_i + \frac{h}{2} (f_{\text{current}} + f_{\text{predicted}}).$$

Example

Solve the IVP:

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

using **Midpoint Method** and **Modified Euler Method** with $N = 10$ steps →
 $h = \frac{2-0}{10} = 0.2$. Exact solution (for error analysis): $y(t) = (t+1)^2 - 0.5e^t$

Step-by-Step Manual Computation (First Two Steps)

Let $f(t, y) = y - t^2 + 1$, $h = 0.2$, $t_0 = 0$, $y_0 = 0.5$.

Midpoint Method

Update rule:

$$k_1 = h f(t_i, y_i), \\ k_2 = h f\left(t_i + \frac{h}{2}, y_i + \frac{k_1}{2}\right), \\ y_{i+1} = y_i + k_2.$$

Step 0 → 1

- $k_1 = 0.2 \cdot f(0, 0.5) = 0.2 \cdot (0.5 - 0 + 1) = 0.3$
- $k_2 = 0.2 \cdot f(0.1, 0.65) = 0.2 \cdot (0.65 - 0.01 + 1) = 0.2 \cdot 1.64 = 0.328$
- $y_1 = 0.5 + 0.328 = \mathbf{0.828}$

Step 1 → 2

- $k_1 = 0.2 \cdot f(0.2, 0.828) = 0.2 \cdot (0.828 - 0.04 + 1) = 0.3576$
- $k_2 = 0.2 \cdot f(0.3, 1.0068) = 0.2 \cdot (1.0068 - 0.09 + 1) = 0.38336$
- $y_2 = 0.828 + 0.38336 = \mathbf{1.21136}$

Modified Euler (Heun's) Method

Update rule:

$$\begin{aligned} k_1 &= h f(t_i, y_i), \\ k_2 &= h f(t_i + h, y_i + k_1), \\ y_{i+1} &= y_i + \frac{1}{2}(k_1 + k_2). \end{aligned}$$

Step 0 → 1

- $k_1 = 0.2 \cdot f(0, 0.5) = 0.3$
- $k_2 = 0.2 \cdot f(0.2, 0.8) = 0.2 \cdot (0.8 - 0.04 + 1) = 0.352$
- $y_1 = 0.5 + \frac{1}{2}(0.3 + 0.352) = \mathbf{0.826}$

Step 1 → 2

- $k_1 = 0.2 \cdot f(0.2, 0.826) = 0.3572$
- $k_2 = 0.2 \cdot f(0.4, 1.1832) = 0.2 \cdot (1.1832 - 0.16 + 1) = 0.2 \cdot 2.0232 = 0.40464$
- $y_2 = 0.826 + \frac{1}{2}(0.3572 + 0.40464) = \mathbf{1.20692}$

```
In [6]: import numpy as np
import matplotlib.pyplot as plt
# Define the ODE RHS
def f(t, y):
    return y - t**2 + 1
# Exact solution: y(t) = (t + 1)^2 - 0.5 * exp(t)
def y_exact(t):
    return (t + 1)**2 - 0.5 * np.exp(t)
# ----- Numerical Methods -----

def euler_method(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t = np.linspace(t0, t_end, N + 1)
    y = np.zeros(N + 1)
    y[0] = y0
    for i in range(N):
        y[i + 1] = y[i] + h * f(t[i], y[i])
    return t, y

def midpoint_method(f, t0, y0, t_end, N):
```

```

h = (t_end - t0) / N
t = np.linspace(t0, t_end, N + 1)
y = np.zeros(N + 1)
y[0] = y0
for i in range(N):
    k1 = h * f(t[i], y[i])
    k2 = h * f(t[i] + h/2, y[i] + k1/2)
    y[i + 1] = y[i] + k2
return t, y

def modified_euler_method(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t = np.linspace(t0, t_end, N + 1)
    y = np.zeros(N + 1)
    y[0] = y0
    for i in range(N):
        k1 = h * f(t[i], y[i])
        k2 = h * f(t[i] + h, y[i] + k1)
        y[i + 1] = y[i] + 0.5 * (k1 + k2)
    return t, y

# ----- Run all methods -----
t0, y0, t_end, N = 0, 0.5, 2, 10
t_eul, y_eul = euler_method(f, t0, y0, t_end, N)
t_mid, y_mid = midpoint_method(f, t0, y0, t_end, N)
t_mod, y_mod = modified_euler_method(f, t0, y0, t_end, N)

# Use t_eul as reference grid (all are identical since same N)
y_ex = y_exact(t_eul)
# Compute absolute errors
err_eul = np.abs(y_eul - y_ex)
err_mid = np.abs(y_mid - y_ex)
err_mod = np.abs(y_mod - y_ex)

# ----- Print comparison table -----
print(f"{'i':>2} {'t_i':>5} {'Euler':>10} {'Midpoint':>10} {'ModEuler':>10} {'Exact'
      f"{'|e_E|':>8} {'|e_M|':>8} {'|e_ME|':>8}")
print("-" * 84)

for i in range(6): # first 6 steps
    print(f"{i:2d} {t_eul[i]:5.1f} "
          f"{y_eul[i]:10.6f} {y_mid[i]:10.6f} {y_mod[i]:10.6f} {y_ex[i]:10.6f} "
          f"{err_eul[i]:8.2e} {err_mid[i]:8.2e} {err_mod[i]:8.2e}")

# Final errors
print("..." + " " * 76)
i = N
print(f"{i:2d} {t_eul[i]:5.1f} "
      f"{y_eul[i]:10.6f} {y_mid[i]:10.6f} {y_mod[i]:10.6f} {y_ex[i]:10.6f} "
      f"{err_eul[i]:8.2e} {err_mid[i]:8.2e} {err_mod[i]:8.2e}")

```

i	t_i	Euler	Midpoint	ModEuler	Exact	e_E	e_M	e_ME
0	0.0	0.500000	0.500000	0.500000	0.500000	0.00e+00	0.00e+00	0.00e+00
1	0.2	0.800000	0.828000	0.826000	0.829299	2.93e-02	1.30e-03	3.30e-03
2	0.4	1.152000	1.211360	1.206920	1.214088	6.21e-02	2.73e-03	7.17e-03
3	0.6	1.550400	1.644659	1.637242	1.648941	9.85e-02	4.28e-03	1.17e-02
4	0.8	1.988480	2.121284	2.110236	2.127230	1.39e-01	5.95e-03	1.70e-02
5	1.0	2.458176	2.633167	2.617688	2.640859	1.83e-01	7.69e-03	2.32e-02
...								
10	2.0	4.865785	5.290369	5.233055	5.305472	4.40e-01	1.51e-02	7.24e-02

```
In [7]: # ----- Plot results -----
plt.figure(figsize=(10, 6))
plt.plot(t_eul, y_ex, 'k-', label='Exact', linewidth=2.5)

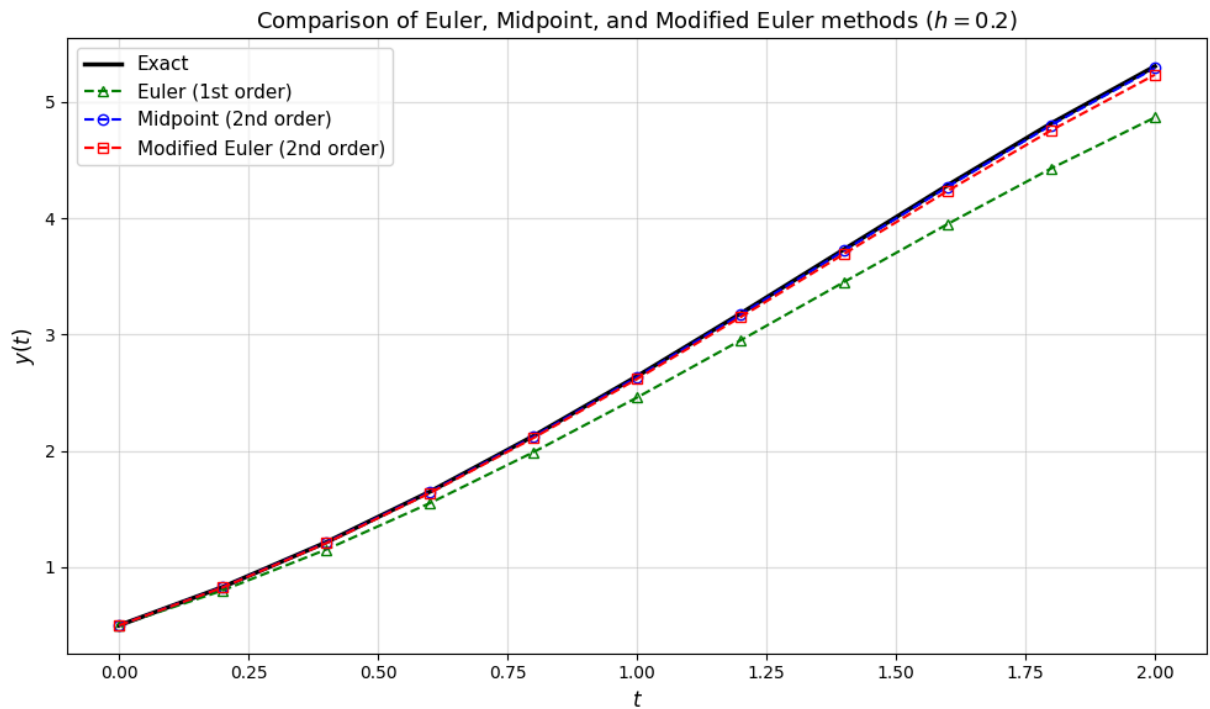
plt.plot(t_eul, y_eul, 'g^--', label='Euler (1st order)', markersize=6, markerfacec
plt.plot(t_mid, y_mid, 'bo--', label='Midpoint (2nd order)', markersize=6, markerfa
plt.plot(t_mod, y_mod, 'rs--', label='Modified Euler (2nd order)', markersize=6, ma

plt.xlabel('$t$', fontsize=12)
plt.ylabel('$y(t)$', fontsize=12)
plt.title('Comparison of Euler, Midpoint, and Modified Euler methods ($h = 0.2$)',
plt.legend(fontsize=11)
plt.grid(True, alpha=0.4)
plt.tight_layout()
plt.show()

# ----- Optional: Convergence check (global error vs h) -----
hs = [0.5, 0.25, 0.125, 0.0625]
errors = {'Euler': [], 'Midpoint': [], 'ModEuler': []}

for h in hs:
    N = int(2 / h)
    _, y_e = euler_method(f, 0, 0.5, 2, N)
    _, y_m = midpoint_method(f, 0, 0.5, 2, N)
    _, y_me = modified_euler_method(f, 0, 0.5, 2, N)
    exact = y_exact(2)

    errors['Euler'].append(abs(y_e[-1] - exact))
    errors['Midpoint'].append(abs(y_m[-1] - exact))
    errors['ModEuler'].append(abs(y_me[-1] - exact))
# Estimate order: Log(error2/error1) / Log(h2/h1)
print("\n Estimated convergence orders (final-time global error):")
for name in errors:
    e = np.array(errors[name])
    orders = np.log(e[:-1] / e[1:]) / np.log(np.array(hs)[:-1] / np.array(hs)[1:])
    print(f"{name:12}: orders ≈ {orders.round(2)} (expected: {1 if name=='Euler' e
```



Estimated convergence orders (final-time global error):

Euler	: orders $\approx [0.72 \ 0.83 \ 0.91]$	(expected: 1)
Midpoint	: orders $\approx [1.94 \ 2. \ 2.01]$	(expected: 2)
ModEuler	: orders $\approx [1.82 \ 1.92 \ 1.96]$	(expected: 2)

Higher-Order Runge–Kutta Methods

Runge–Kutta Method of Third Order (Kutta’s RK3)

We derive **Kutta’s third-order Runge–Kutta method** by enforcing third-order accuracy on a general 3-stage explicit RK scheme.

A 3-stage explicit RK method has the form:

$$\begin{cases} k_1 = f(t_i, y_i), \\ k_2 = f(t_i + c_2 h, y_i + h a_{21} k_1), \\ k_3 = f(t_i + c_3 h, y_i + h (a_{31} k_1 + a_{32} k_2)), \\ y_{i+1} = y_i + h (b_1 k_1 + b_2 k_2 + b_3 k_3). \end{cases}$$

Explicitness requires:

$$a_{21} = c_2, \quad c_3 = a_{31} + a_{32}.$$

To achieve **local truncation error** $\mathcal{O}(h^4)$ (i.e., **global error** $\mathcal{O}(h^3)$), the coefficients must satisfy the **four classical order conditions**:

Condition

Equation

(C1)

Condition	Equation
	$ \begin{aligned} &b_1 + b_2 \\ &+ b_3 \\ &= 1 \end{aligned} $
(C2)	$ \begin{aligned} &b_2 c_2 \\ &+ b_3 c_3 \\ &= \frac{1}{2} \end{aligned} $
(C3)	$ \begin{aligned} &b_2 c_2^2 \\ &+ b_3 c_3^2 \\ &= \frac{1}{3} \end{aligned} $
(C4)	$ \begin{aligned} &b_3 a_{32} c_2 \\ &= \frac{1}{6} \end{aligned} $

These ensure the numerical update matches the Taylor expansion of the exact solution up to $\mathcal{O}(h^3)$.

Kutta (1901) chose natural stage points in $[t_i, t_{i+1}]$:

$$c_2 = \frac{1}{2}, \quad c_3 = 1,$$

corresponding to t_i , $t_i + h/2$, and $t_i + h$ — analogous to Simpson's rule.

Step 1: Solve for b_1, b_2, b_3

From (C2) and (C3):

$$\begin{cases} \frac{1}{2}b_2 + b_3 = \frac{1}{2} & \text{(C2)} \\ \frac{1}{4}b_2 + b_3 = \frac{1}{3} & \text{(C3)} \end{cases} \Rightarrow b_3 = \frac{1}{6}, \quad b_2 = \frac{2}{3}, \quad b_1 = 1 - b_2 - b_3 = \frac{1}{6}.$$

Step 2: Solve for a_{32}, a_{31}

From (C4):

$$b_3 a_{32} c_2 = \frac{1}{6} \Rightarrow \frac{1}{6} \cdot a_{32} \cdot \frac{1}{2} = \frac{1}{6} \Rightarrow a_{32} = 2.$$

From consistency $c_3 = a_{31} + a_{32}$:

$$1 = a_{31} + 2 \Rightarrow a_{31} = -1.$$

Kutta's Third-Order Method

Substituting all coefficients gives:

$$\begin{cases} k_1 = f(t_i, y_i), \\ k_2 = f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right), \\ k_3 = f(t_i + h, y_i - hk_1 + 2hk_2), \\ y_{i+1} = y_i + \frac{h}{6}(k_1 + 4k_2 + k_3). \end{cases}$$

This resembles Simpson's rule: weights 1–4–1 over three points.

Butcher Tableau

$$\begin{array}{c|ccc} 0 & & & \\ \frac{1}{2} & \frac{1}{2} & & \\ 1 & -1 & 2 & \\ \hline & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{array}$$

Verification of Order Conditions

Condition	LHS Computation	Required
(C1) $b_1 + b_2 + b_3$	$\frac{1}{6} + \frac{2}{3} + \frac{1}{6} = 1$	1
(C2) $b_2c_2 + b_3c_3$	$\frac{2}{3} \cdot \frac{1}{2} + \frac{1}{6} \cdot 1 = \frac{1}{2}$	$\frac{1}{2}$
(C3) $b_2c_2^2 + b_3c_3^2$	$\frac{2}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot 1 = \frac{1}{3}$	$\frac{1}{3}$
(C4) $b_3a_{32}c_2$	$\frac{1}{6} \cdot 2 \cdot \frac{1}{2} = \frac{1}{6}$	$\frac{1}{6}$

All conditions satisfied → **local error** $\mathcal{O}(h^4)$, **global error** $\mathcal{O}(h^3)$.

Runge–Kutta Method of Order Four (Classical RK4)

The most widely used Runge–Kutta method is the **classical fourth-order scheme** (RK4), defined by:

$$\left\{ \begin{array}{l} w_0 = \alpha, \\ k_1 = h f(t_i, w_i), \\ k_2 = h f\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_1\right), \\ k_3 = h f\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_2\right), \\ k_4 = h f(t_i + h, w_i + k_3), \\ w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4), \end{array} \right. \quad i = 0, 1, \dots, N-1.$$

Properties:

- Local truncation error: $\mathcal{O}(h^5)$
- Global error: $\mathcal{O}(h^4)$
- Requires **4 evaluations of f per step**
- No derivatives needed — fully explicit and robust

Butcher Tableau for Classical RK4

0				
$\frac{1}{2}$	$\frac{1}{2}$			
$\frac{1}{2}$	0	$\frac{1}{2}$		
1	0	0	1	
	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{6}$

Weights $\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6}$ — symmetric and reminiscent of Simpson's rule.

Example

Use the **classical fourth-order Runge–Kutta method** with

- step size $h = 0.2$,
- $N = 10$ steps over $[0, 2]$,
- nodes $t_i = 0.2 i$ to solve the IVP $y' = y - t^2 + 1, y(0) = 0.5$ Exact solution:

$$y(t) = (t+1)^2 - \frac{1}{2}e^t$$

Manual Computation: First Step ($t_0 = 0 \rightarrow t_1 = 0.2$)

Let $f(t, y) = y - t^2 + 1, w_0 = y(0) = 0.5, h = 0.2$.

$$K_1 = h \cdot f(t_0, w_0) = 0.2 \cdot f(0, 0.5) = 0.2 \cdot (0.5 - 0 + 1) = 0.2 \cdot 1.5 = \mathbf{0.300000}$$

$$\begin{aligned} K_2 &= h \cdot f\left(t_0 + \frac{h}{2}, w_0 + \frac{K_1}{2}\right) = 0.2 \cdot f(0.1, 0.65) \\ &= 0.2 \cdot (0.65 - 0.1^2 + 1) = 0.2 \cdot (0.65 - 0.01 + 1) = 0.2 \cdot 1.64 = \mathbf{0.328000} \end{aligned}$$

$$\begin{aligned} K_3 &= h \cdot f\left(t_0 + \frac{h}{2}, w_0 + \frac{K_2}{2}\right) = 0.2 \cdot f(0.1, 0.5 + 0.328/2) = 0.2 \cdot f(0.1, 0.664) \\ &= 0.2 \cdot (0.664 - 0.01 + 1) = 0.2 \cdot 1.654 = \mathbf{0.330800} \end{aligned}$$

$$\begin{aligned} K_4 &= h \cdot f(t_0 + h, w_0 + K_3) = 0.2 \cdot f(0.2, 0.5 + 0.3308) = 0.2 \cdot f(0.2, 0.8308) \\ &= 0.2 \cdot (0.8308 - 0.2^2 + 1) = 0.2 \cdot (0.8308 - 0.04 + 1) = 0.2 \cdot 1.7908 = \mathbf{0.358160} \end{aligned}$$

Update:

$$\begin{aligned} w_1 &= w_0 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4) \\ &= 0.5 + \frac{1}{6}(0.300000 + 2(0.328000) + 2(0.330800) + 0.358160) \\ &= 0.5 + \frac{1.975760}{6} = 0.5 + 0.3292933\dots = \mathbf{0.829293} \end{aligned}$$

```
In [8]: import numpy as np
import matplotlib.pyplot as plt

# Define ODE and exact solution
def f(t, y):
    return y - t**2 + 1

def y_exact(t):
    return (t + 1)**2 - 0.5 * np.exp(t)

# Classical RK4 method
def rk4_step(f, t, y, h):
    k1 = h * f(t, y)
    k2 = h * f(t + h/2, y + k1/2)
    k3 = h * f(t + h/2, y + k2/2)
    k4 = h * f(t + h, y + k3)
    return y + (k1 + 2*k2 + 2*k3 + k4) / 6

def rk4_full(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t_vals = [t0]
    y_vals = [y0]
    y = y0
    t = t0
    for i in range(N):
        y = rk4_step(f, t, y, h)
        t += h
        t_vals.append(t)
        y_vals.append(y)
    return np.array(t_vals), np.array(y_vals)

# Run RK4
t0, y0, t_end, N = 0, 0.5, 2, 10
```

```

t_rk4, w_rk4 = rk4_full(f, t0, y0, t_end, N)
y_ex = y_exact(t_rk4)
abs_err = np.abs(w_rk4 - y_ex)

# Print table manually (no pandas)
print(f"{'i':>2} {'t_i':>4} {'w_i (RK4)':>12} {'y(t_i)':>12} {'|error|':>10}")
print("-" * 48)
for i in range(len(t_rk4)):
    print(f"{'i':2d} {'t_rk4[i]:4.1f} {'w_rk4[i]:12.6f} {'y_ex[i]:12.6f} {'abs_err[i]:10.6f}")

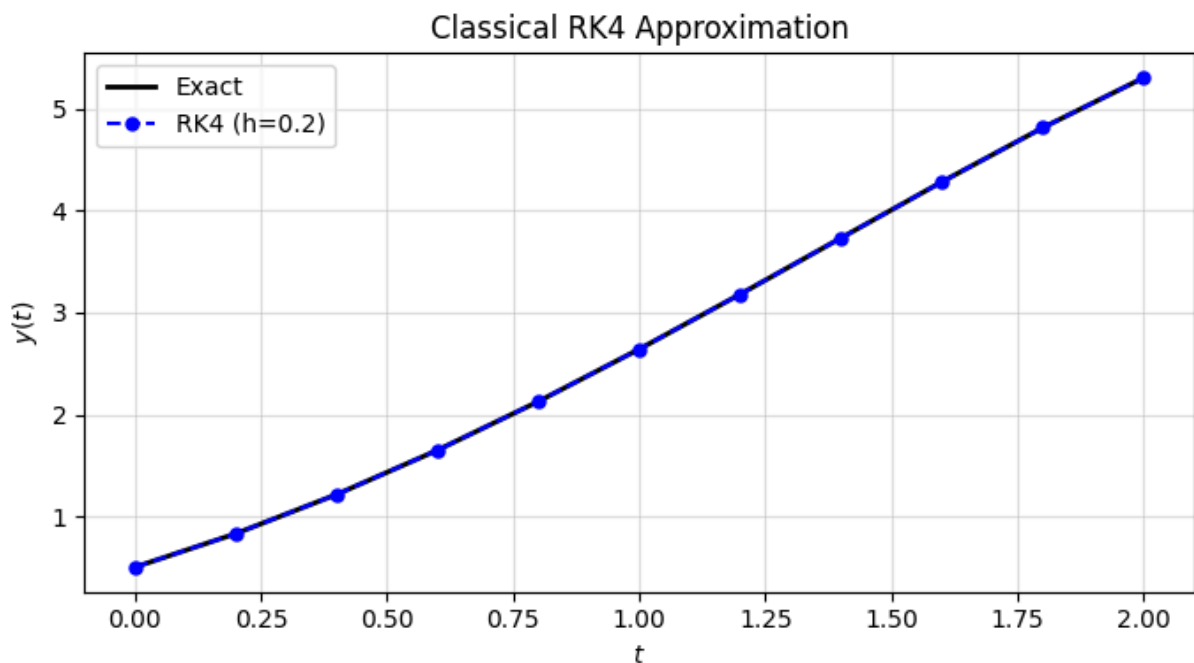
```

i	t_i	w_i (RK4)	y(t_i)	error
0	0.0	0.500000	0.500000	0.00e+00
1	0.2	0.829293	0.829299	5.29e-06
2	0.4	1.214076	1.214088	1.14e-05
3	0.6	1.648922	1.648941	1.86e-05
4	0.8	2.127203	2.127230	2.69e-05
5	1.0	2.640823	2.640859	3.64e-05
6	1.2	3.179894	3.179942	4.74e-05
7	1.4	3.732340	3.732400	5.99e-05
8	1.6	4.283409	4.283484	7.43e-05
9	1.8	4.815086	4.815176	9.06e-05
10	2.0	5.305363	5.305472	1.09e-04

```

In [9]: plt.figure(figsize=(7, 4))
plt.plot(t_rk4, y_ex, 'k-', label='Exact', linewidth=2)
plt.plot(t_rk4, w_rk4, 'bo--', label='RK4 (h=0.2)', markersize=5)
plt.xlabel('$t$'); plt.ylabel('$y(t)$')
plt.title('Classical RK4 Approximation')
plt.legend(); plt.grid(True, alpha=0.4)
plt.tight_layout()
plt.show()

```



Why RK4 Dominates Practice

- High accuracy with modest h
- No derivative computations (unlike Taylor methods)
- Excellent stability for non-stiff problems
- Symmetric weighting improves error cancellation

Note: For stiff problems, implicit methods (e.g., backward Euler, BDF) are preferred — but RK4 remains the **workhorse for non-stiff IVPs** in science and engineering.

Computational Comparisons

The main computational cost in Runge–Kutta methods comes from **evaluations of $f(t, y)$** per step.

- **Second-order RK methods** (e.g., Midpoint, Modified Euler):
→ 2 function evaluations per step, local truncation error $\mathcal{O}(h^3)$ (global $\mathcal{O}(h^2)$).
- **Classical RK4:**
→ 4 evaluations per step, local truncation error $\mathcal{O}(h^5)$ (global $\mathcal{O}(h^4)$).

Butcher’s theory establishes the optimal relationship between the number of stages (function evaluations) and achievable order:

Best Possible Local Truncation Error vs. Function Evaluations

Evaluations per step	2	3	4	5–7	≥ 8
Best possible local truncation error	$\mathcal{O}(h^3)$	$\mathcal{O}(h^4)$	$\mathcal{O}(h^5)$	$\mathcal{O}(h^n)$	$\mathcal{O}(h^{n-1})$

 **Note:** Local truncation error for an m -stage method is generally $\mathcal{O}(h^{p+1})$, where p is the **global order**.
Common convention: A “fourth-order RK method” has global error $\mathcal{O}(h^4)$, so local error $\mathcal{O}(h^5)$.

Key Insight:

- For $n = 5, 6, 7$ stages, the maximal **global** order is $n - 1$ (local error $\mathcal{O}(h^n)$).
- For $n \geq 8$, the maximal global order drops to $n - 2$ (local error $\mathcal{O}(h^{n-1})$).

Hence, **RK4 (4 stages, 4th order)** is near-optimal: it achieves the theoretical maximum global order for its cost. This explains its dominance in practice over higher-stage methods for non-stiff problems.

Practical takeaway: Smaller step sizes with moderate-order methods (e.g., RK4) often outperform large-step high-order methods due to better stability, simplicity, and error control.

Example: For Fair Computational Comparison

We consider the initial value problem:

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

To ensure a **fair comparison** of numerical methods, we fix the **total number of function evaluations** (i.e., calls to $f(t, y)$) across all methods — since this dominates computational cost.

This allows us to assess *accuracy per unit work*, rather than per step.

Computational Comparison with Equal Function Evaluations

We solve the IVP:

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

using three methods, each with **exactly 40 evaluations of $f(t, y)$** :

Method	Evaluations/step	Step size h	Steps N	Total evals
Euler	1	0.05	40	40
Modified Euler	2	0.10	20	40
RK4	4	0.20	10	40

Fair comparison: same computational cost, different accuracy.

```
In [10]: import numpy as np

# Define ODE and exact solution
def f(t, y):
    return y - t**2 + 1

def y_exact(t):
    return (t + 1)**2 - 0.5 * np.exp(t)

# --- Euler (1 eval/step) ---
def euler(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t = np.linspace(t0, t_end, N+1)
    y = np.zeros(N+1)
    y[0] = y0
    for i in range(N):
        y[i+1] = y[i] + h * f(t[i], y[i])
```

```

    return t, y
# --- Modified Euler (2 evals/step) ---
def modified_euler(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t = np.linspace(t0, t_end, N+1)
    y = np.zeros(N+1)
    y[0] = y0
    for i in range(N):
        k1 = h * f(t[i], y[i])
        k2 = h * f(t[i] + h, y[i] + k1)
        y[i+1] = y[i] + 0.5 * (k1 + k2)
    return t, y
# --- RK4 (4 evals/step) ---
def rk4(f, t0, y0, t_end, N):
    h = (t_end - t0) / N
    t = np.linspace(t0, t_end, N+1)
    y = np.zeros(N+1)
    y[0] = y0
    for i in range(N):
        ti, yi = t[i], y[i]
        k1 = h * f(ti, yi)
        k2 = h * f(ti + h/2, yi + k1/2)
        k3 = h * f(ti + h/2, yi + k2/2)
        k4 = h * f(ti + h, yi + k3)
        y[i+1] = yi + (k1 + 2*k2 + 2*k3 + k4) / 6
    return t, y

# Run all methods with 40 f-evaluations
t0, y0, t_end = 0, 0.5, 2

# Euler: h = 0.05, N = 40
t_e, y_e = euler(f, t0, y0, t_end, 40)
# Modified Euler: h = 0.1, N = 20
t_m, y_m = modified_euler(f, t0, y0, t_end, 20)
# RK4: h = 0.2, N = 10
t_r, y_r = rk4(f, t0, y0, t_end, 10)

# Exact values at all points (interpolate for consistency)
t_plot = np.array([0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0])
y_ex = y_exact(t_plot)

# Extract approximations at t_plot
y_e_interp = np.interp(t_plot, t_e, y_e) # Euler
y_m_interp = np.interp(t_plot, t_m, y_m) # Modified Euler
y_r_interp = np.interp(t_plot, t_r, y_r) # RK4

# Compute absolute errors
err_e = np.abs(y_e_interp - y_ex)
err_m = np.abs(y_m_interp - y_ex)
err_r = np.abs(y_r_interp - y_ex)

# Header
print(f"{'t_i':>4} {'Exact':>12} "
      f"{'Euler':>10} {'|e|':>8} "
      f"{'ModEuler':>10} {'|e|':>8} "
      f"{'RK4':>10} {'|e|':>8}")

```

```
print("-" * 74)
```

```
# Rows
```

```
for i, ti in enumerate(t_plot):
    print(f"ti:4.1f} {y_ex[i]:12.7f} "
          f"{y_e_interp[i]:10.7f} {err_e[i]:8.2e} "
          f"{y_m_interp[i]:10.7f} {err_m[i]:8.2e} "
          f"{y_r_interp[i]:10.7f} {err_r[i]:8.2e}")
```

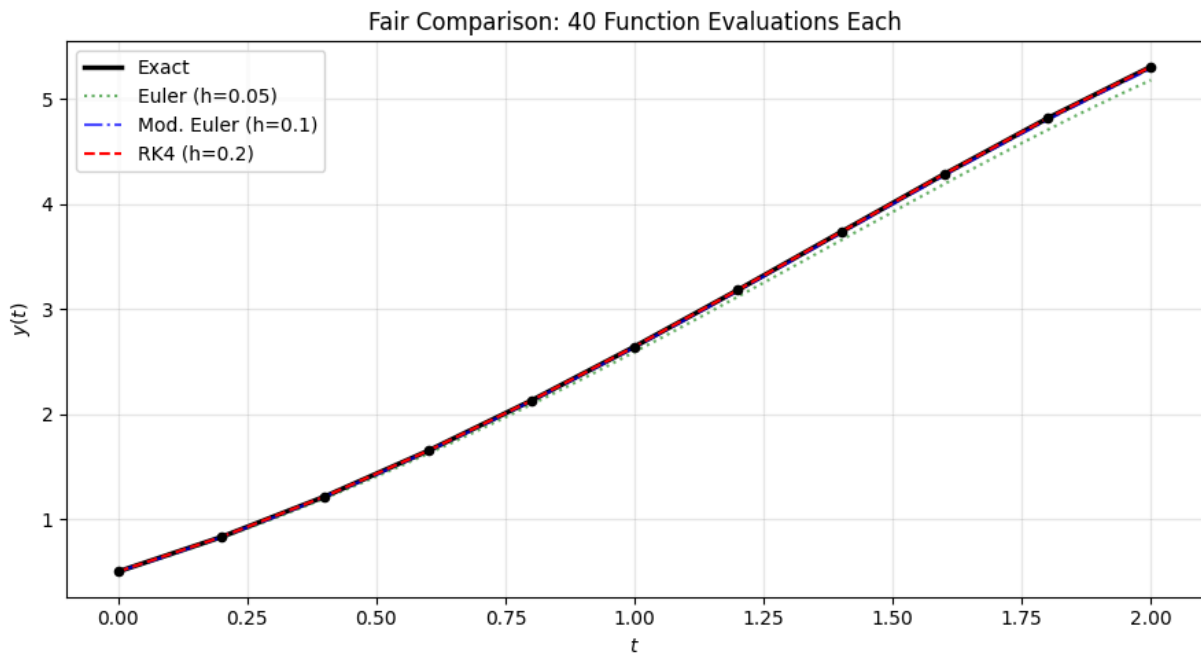
t_i	Exact	Euler	e	ModEuler	e	RK4	e
0.0	0.5000000	0.5000000	0.00e+00	0.5000000	0.00e+00	0.5000000	0.00e+00
0.2	0.8292986	0.8214716	7.83e-03	0.8284350	8.64e-04	0.8292933	5.29e-06
0.4	1.2140877	1.1973995	1.67e-02	1.2122113	1.88e-03	1.2140762	1.14e-05
0.6	1.6489406	1.6222790	2.67e-02	1.6458789	3.06e-03	1.6489220	1.86e-05
0.8	2.1272295	2.0894190	3.78e-02	2.1227827	4.45e-03	2.1272027	2.69e-05
1.0	2.6408591	2.5906863	5.02e-02	2.6347973	6.06e-03	2.6408227	3.64e-05
1.2	3.1799415	3.1161950	6.37e-02	3.1720009	7.94e-03	3.1798942	4.74e-05
1.4	3.7324000	3.6539290	7.85e-02	3.7222788	1.01e-02	3.7323401	5.99e-05
1.6	4.2834838	4.1892822	9.42e-02	4.2708390	1.26e-02	4.2834095	7.43e-05
1.8	4.8151763	4.7045011	1.11e-01	4.7996197	1.56e-02	4.8150857	9.06e-05
2.0	5.3054720	5.1780062	1.27e-01	5.2865672	1.89e-02	5.3053630	1.09e-04

In [11]: `import matplotlib.pyplot as plt`

```
plt.figure(figsize=(9, 5))
plt.plot(t_plot, y_ex, 'k-', label='Exact', linewidth=2.5)
plt.plot(t_e, y_e, 'g:', alpha=0.6, label='Euler (h=0.05)')
plt.plot(t_m, y_m, 'b-.', alpha=0.7, label='Mod. Euler (h=0.1)')
plt.plot(t_r, y_r, 'r--', label='RK4 (h=0.2)')
plt.scatter(t_plot, y_ex, c='k', s=20, zorder=5)
plt.xlabel('$t$'); plt.ylabel('$y(t)$')
plt.title('Fair Comparison: 40 Function Evaluations Each')
plt.legend(); plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
# Final errors
```

```
print("\n✅ Final errors at t = 2.0:")
print(f"Euler (1st order)      : {err_e[-1]:.2e}")
print(f"Modified Euler (2nd)   : {err_m[-1]:.2e}")
print(f"RK4 (4th order)         : {err_r[-1]:.2e}")
```



✓ Final errors at $t = 2.0$:

Euler (1st order) : $1.27\text{e-}01$
 Modified Euler (2nd) : $1.89\text{e-}02$
 RK4 (4th order) : $1.09\text{e-}04$

Lab Work Exercise: Use Runge–Kutta Methods of Order 2, 3, and 4

Implement and compare the following methods:

- **Order 2:** Modified Euler and Midpoint
- **Order 3:** Kutta's RK3
- **Order 4:** Classical RK4

for each IVP below. Use the indicated step size h , and compare numerical results with the exact solution.

a. $y' = \frac{2-2ty}{t^2+1}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$ Exact solution: $y(t) = \frac{2t+1}{t^2+1}$

b. $y' = \frac{y^2}{1+t}$, $1 \leq t \leq 2$, $y(1) = -\frac{1}{\ln 2}$, $h = 0.1$ Exact solution: $y(t) = -\frac{1}{\ln(1+t)}$

c. $y' = \frac{y^2+y}{t}$, $1 \leq t \leq 3$, $y(1) = -2$, $h = 0.2$ Exact solution: $y(t) = \frac{2t}{1-2t}$

d. $y' = -ty + \frac{4t}{y}$, $0 \leq t \leq 1$, $y(0) = 1$, $h = 0.1$ Exact solution:
 $y(t) = \sqrt{4 - 3e^{-t^2}}$

Results must contain:

- Tables of t_i , numerical w_i , exact $y(t_i)$, and absolute error for each method.
- Plots of solutions and error curves.
- Brief discussion: Which method gives best accuracy for given h ? How do errors scale with order?

Stability Analysis of Single-Step Explicit Methods

Numerical methods for IVPs (e.g., Euler, Runge–Kutta) must balance **accuracy**, **convergence**, and **stability**.

While convergence tells us the method approaches the true solution as $h \rightarrow 0$, **stability** ensures errors do not grow uncontrollably during integration.

We analyze stability using the **linear test equation**:

$$y' = \lambda y, \quad y(0) = y_0,$$

with $\lambda \in \mathbb{C}$. The exact solution is $y(t) = y_0 e^{\lambda t}$.

- If $\Re(\lambda) < 0$: solution decays (stable).
- A numerical method is **absolutely stable** if its solution mimics this decay for a given step size h .

Define $z = h\lambda$. The behavior of the numerical solution is governed by the **stability function** $R(z)$, where:

$$y_{n+1} = R(z) y_n.$$

The **region of absolute stability** is:

$$\mathcal{S} = \{z \in \mathbb{C} \mid |R(z)| \leq 1\}.$$

◆ Euler's Method (RK1)

Update:

$$y_{n+1} = y_n + hf(t_n, y_n)$$

For $f(t, y) = \lambda y$:

$$y_{n+1} = y_n + h\lambda y_n = (1 + z)y_n, \quad z = h\lambda.$$

✓ **Stability function:**

$$R(z) = 1 + z$$

✓ **Stability region:**

$$|1 + z| \leq 1$$

→ Disk of radius 1 centered at $z = -1$ in the complex plane.

★ **Real-axis implication** ($\lambda < 0$):

Stability requires

$$-2 \leq z = h\lambda < 0 \quad \Rightarrow \quad 0 < h \leq \frac{2}{|\lambda|}.$$

→ Step size limited for stiff problems.

◆ Second-Order Runge–Kutta (Midpoint Method)

Scheme:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\ y_{n+1} &= y_n + h k_2. \end{aligned}$$

Apply to $y' = \lambda y$ ($f = \lambda y$):

$$\begin{aligned} k_1 &= \lambda y_n, \\ k_2 &= \lambda\left(y_n + \frac{h}{2}\lambda y_n\right) = \lambda y_n\left(1 + \frac{z}{2}\right), \\ y_{n+1} &= y_n + h \cdot \lambda y_n\left(1 + \frac{z}{2}\right) = y_n\left(1 + z + \frac{z^2}{2}\right). \end{aligned}$$

✓ **Stability function:**

$$R(z) = 1 + z + \frac{z^2}{2}$$

✓ **Stability region:** $|1 + z + z^2/2| \leq 1$

→ Larger than Euler's; contains real interval $[-2, 0]$.

◆ Third-Order Runge–Kutta (Kutta's RK3)

Scheme:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \\ k_3 &= f(t_n + h, y_n - hk_1 + 2hk_2), \\ y_{n+1} &= y_n + h\left(\frac{1}{6}k_1 + \frac{2}{3}k_2 + \frac{1}{6}k_3\right). \end{aligned}$$

Apply to $f = \lambda y$, $z = h\lambda$:

$$\begin{aligned} k_1 &= \lambda y_n, \\ k_2 &= \lambda y_n\left(1 + \frac{z}{2}\right), \\ k_3 &= \lambda y_n(1 + z + z^2), \\ y_{n+1} &= y_n\left(1 + z + \frac{z^2}{2} + \frac{z^3}{6}\right). \end{aligned}$$

✓ **Stability function:**

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}$$

✚ This is the **third-order Taylor polynomial** of e^z — as expected for a 3rd-order method.

◆ Stability Functions Summary

Method	Order	Stability Function $R(z)$
Euler (RK1)	1	$1 + z$
Midpoint (RK2)	2	$1 + z + \frac{z^2}{2}$
Kutta's RK3	3	$1 + z + \frac{z^2}{2} + \frac{z^3}{6}$
Classical RK4	4	$1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$

🎯 Key observations:

- $R(z)$ matches the Taylor expansion of e^z to order p for a p -th order method.
- All **explicit** RK methods have **bounded** stability regions.
- Stability region grows with order $\rightarrow$ larger stable h for non-stiff problems.
- For **stiff** problems ($\Re(\lambda) \ll 0$), implicit methods (e.g., backward Euler, Radau) are required — their stability regions include the entire left-half plane (A-stable or L-stable).

📊 Stability Regions (Real Axis Approximation)

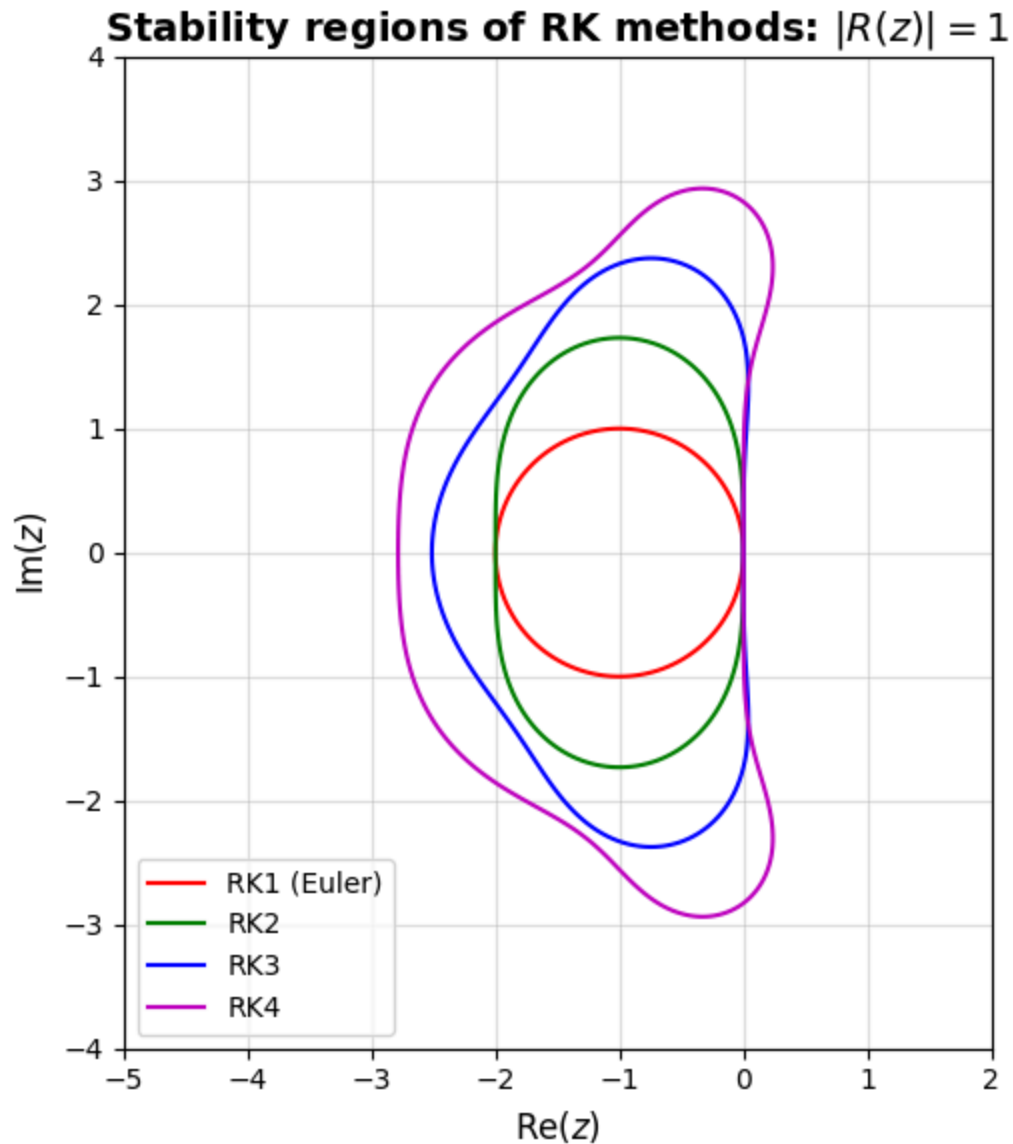
On the negative real axis ($z < 0$), the interval of absolute stability is approximately:

Method	Stable z interval
Euler (RK1)	$[-2, 0]$
RK2	$[-2.0, 0]$ (exact: $z \in [-2, 0]$)
RK3	$[-2.51, 0]$
RK4	$[-2.79, 0]$

$\rightarrow$ Each increase in order allows ~20–30% larger stable step size for real negative eigenvalues.

⚠ But for complex λ (e.g., oscillatory systems), the full 2D region matters — explicit methods still fail for large $|\Re(\lambda)|$.

```
In [12]: import numpy as np
import matplotlib.pyplot as plt
# Define complex grid
x = np.linspace(-5, 2, 500)
y = np.linspace(-4, 4, 500)
X, Y = np.meshgrid(x, y)
Z = X + 1j * Y
# Stability functions  $R(z)$  = Taylor expansion of  $\exp(z)$ 
R1 = 1 + Z # RK1 (Euler)
R2 = 1 + Z + Z**2 / 2 # RK2
R3 = 1 + Z + Z**2 / 2 + Z**3 / 6 # RK3
R4 = 1 + Z + Z**2 / 2 + Z**3 / 6 + Z**4 / 24 # RK4
# Magnitudes
modR1 = np.abs(R1)
modR2 = np.abs(R2)
modR3 = np.abs(R3)
modR4 = np.abs(R4)
# Plot
plt.figure(figsize=(8, 6))
# Contours where  $|R(z)| = 1$ 
plt.contour(X, Y, modR1, levels=[1], colors='r', linewidths=1.5)
plt.contour(X, Y, modR2, levels=[1], colors='g', linewidths=1.5)
plt.contour(X, Y, modR3, levels=[1], colors='b', linewidths=1.5)
plt.contour(X, Y, modR4, levels=[1], colors='m', linewidths=1.5)
# Formatting
plt.gca().set_aspect('equal')
plt.xlabel('Re($z$)', fontsize=12)
plt.ylabel('Im($z$)', fontsize=12)
plt.title('Stability regions of RK methods:  $|R(z)| = 1$ ', fontsize=14, fontweight='bold')
plt.xlim(-5, 2)
plt.ylim(-4, 4)
plt.grid(True, alpha=0.4)
# Legend with valid location
from matplotlib.lines import Line2D
legend_elements = [
    Line2D([0], [0], color='r', lw=1.5, label='RK1 (Euler)'),
    Line2D([0], [0], color='g', lw=1.5, label='RK2'),
    Line2D([0], [0], color='b', lw=1.5, label='RK3'),
    Line2D([0], [0], color='m', lw=1.5, label='RK4')]
plt.legend(handles=legend_elements, loc='lower left') # ✅ Fixed: 'southeast' → 'lower left'
plt.tight_layout()
plt.show()
```



The stability regions grow with the order of the method, making higher-order Runge--Kutta methods more robust for non-stiff problems. However, for stiff systems, implicit methods are required due to the bounded nature of explicit stability regions.

🔑 Key Observations

- All **explicit Runge--Kutta methods** have **polynomial stability functions** $R(z)$, since they are derived from explicit formulas without solving algebraic equations.
- Their **stability regions are bounded** in the complex plane:

$$\mathcal{S} = \{z \in \mathbb{C} : |R(z)| \leq 1\} \quad \text{is bounded.}$$

They **cannot stably integrate stiff problems** unless the step size h is very small (often impractically so).

- **Higher-order explicit RK methods** have **larger stability regions**, allowing larger stable step sizes for non-stiff or mildly stiff problems — but the improvement is *modest* (e.g., RK4 stable up to $z \approx -2.79$ on the real axis vs. RK1's $z = -2$).
- For **real negative eigenvalues** ($\lambda < 0$), the maximum stable step satisfies:

$$h_{\max} \approx \frac{C_p}{|\lambda|}, \quad \text{where } C_1 = 2, C_2 = 2, C_3 \approx 2.51, C_4 \approx 2.79.$$

Accuracy *and* stability improve with order — but only up to a point.

💡 **Takeaway:** Stability analysis via the test equation $y' = \lambda y$ is a powerful diagnostic tool. While explicit RK methods (Euler → RK4) offer increasing accuracy and modest stability gains, **stiff systems require implicit methods** (e.g., backward Euler, trapezoidal rule, BDF), whose stability regions include the entire left-half plane (A-stable or L-stable).

Exercise

Modify the code to overlay all four stability boundaries on a single plot for comparison to the linear test equation:

$$y' = -\lambda y, \quad y(0) = y_0,$$

where $\lambda \in \mathbb{C}$ and $\Re(\lambda) > 0$ (ensuring the exact solution decays as $t \rightarrow \infty$). The exact solution is $y(t) = y_0 e^{-\lambda t}$, which satisfies $|y(t)| \rightarrow 0$ as $t \rightarrow \infty$ whenever $\Re(\lambda) > 0$.

2.2 Multistep Methods

The numerical methods discussed earlier such as Euler's method and Runge–Kutta methods are called **one-step methods** because the approximation at t_{i+1} depends only on information from the single previous point t_i .

In contrast, **multistep methods** use approximations from *multiple* previous time steps to compute y_{i+1} . Since the solution has already been approximated at $t_0, t_1, \dots, t_i$, it is natural to exploit this historical data to construct higher-order, more efficient schemes.

Definition: Multistep Method

An m -step multistep method for solving the initial value problem

$$y' = f(t, y), \quad y(a) = y_0, \quad a \leq t \leq b,$$

is defined by the recurrence relation

$$y_{i+1} = \sum_{j=0}^{m-1} a_j y_{i-j} + h \sum_{j=0}^m b_j f(t_{i-j+1}, y_{i-j+1}), \quad i = m-1, m, \dots, N-1,$$

where:

- $h = \frac{b-a}{N}$ is the uniform step size
- a_j, b_j are fixed real constants (independent of h)
- $f_k = f(t_k, y_k)$ denotes the derivative at step k

The method is classified as:

- **Explicit** if $b_m = 0$ (so f_{i+1} does not appear)
- **Implicit** if $b_m \neq 0$ (so y_{i+1} appears on both sides and requires solving an equation)

Remark

To start an m -step method, we need m initial values:

$$y_0, y_1, \dots, y_{m-1}.$$

Since only y_0 is given by the initial condition, the remaining $m - 1$ values must be computed using a one-step method (e.g., a fourth-order Runge–Kutta method) of comparable accuracy.

Common Examples of Multistep Methods

Multistep methods are often derived by integrating the ODE over an interval and approximating the integral using polynomial interpolation of f . Two important families are the Adams–Bashforth (explicit) and Adams–Moulton (implicit) methods.

Adams–Bashforth Methods (Explicit)

These are derived by interpolating f at previous points $t_i, t_{i-1}, \dots$ and integrating forward.

- **Two-step AB2:**

$$y_{i+1} = y_i + \frac{h}{2}(3f_i - f_{i-1})$$

- **Three-step AB3:**

$$y_{i+1} = y_i + \frac{h}{12}(23f_i - 16f_{i-1} + 5f_{i-2})$$

- **Four-step AB4:**

$$y_{i+1} = y_i + \frac{h}{24}(55f_i - 59f_{i-1} + 37f_{i-2} - 9f_{i-3})$$

Adams–Moulton Methods (Implicit)

These use interpolation that includes the future point t_{i+1} , leading to implicit formulas with higher accuracy.

- **Two-step AM2 (also called the trapezoidal rule when $m = 1$):**

$$y_{i+1} = y_i + \frac{h}{12} (5f_{i+1} + 8f_i - f_{i-1})$$

- **Three-step AM3:**

$$y_{i+1} = y_i + \frac{h}{24} (9f_{i+1} + 19f_i - 5f_{i-1} + f_{i-2})$$

Derivation via Polynomial Interpolation

Multistep methods are derived by integrating the differential equation

$$y'(t) = f(t, y(t))$$

over a single step:

$$y(t_{i+1}) = y(t_i) + \int_{t_i}^{t_{i+1}} f(t, y(t)) dt.$$

Since the exact solution $y(t)$ is unknown, we approximate the integrand $f(t, y(t))$ by a polynomial $P(t)$ that interpolates known (or previously computed) values of f . The choice of interpolation points determines whether the resulting method is explicit or implicit.

Adams–Bashforth Two-Step Method (AB2)

To obtain an explicit method, we interpolate f at the two most recent *past* points: (t_{i-1}, f_{i-1}) and (t_i, f_i) , where $f_j = f(t_j, y_j)$.

The linear interpolating polynomial is

$$P_1(t) = f_{i-1} \frac{t_i - t}{h} + f_i \frac{t - t_{i-1}}{h}.$$

Integrating $P_1(t)$ from t_i to t_{i+1} gives:

$$\int_{t_i}^{t_{i+1}} P_1(t) dt = \frac{h}{2} (3f_i - f_{i-1}).$$

Substituting into the integral form yields the two-step Adams–Bashforth method:

$$y_{i+1} = y_i + \frac{h}{2} (3f_i - f_{i-1}).$$

This method is explicit (no f_{i+1} appears) and second-order accurate.

Algorithm: Adams–Bashforth 2-Step Method (AB2)

Input:

- Initial value y_0 at $t_0 = a$
- Step size h
- Interval $[a, b]$
- Function $f(t, y)$

Output:

Approximations $y_1, y_2, \dots, y_N$ where $N = (b - a)/h$

Procedure:

1. **Starting value:** Compute y_1 using a one-step method of at least second-order accuracy (e.g., RK2 or RK4) to maintain overall accuracy.

2. **Main loop:** For $i = 1, 2, \dots, N - 1$:

- Evaluate $f_i = f(t_i, y_i)$ and $f_{i-1} = f(t_{i-1}, y_{i-1})$
- Update solution:

$$y_{i+1} = y_i + \frac{h}{2}(3f_i - f_{i-1})$$

3. Return the sequence $\{y_0, y_1, \dots, y_N\}$.

Adams–Moulton Two-Step Method (AM2)

For higher accuracy, we include the future point (t_{i+1}, f_{i+1}) in the interpolation. Using three points t_{i-1}, t_i, t_{i+1} we construct a quadratic interpolating polynomial $P_2(t)$.

Integrating $P_2(t)$ over $[t_i, t_{i+1}]$ results in:

$$\int_{t_i}^{t_{i+1}} P_2(t) dt = \frac{h}{12}(5f_{i+1} + 8f_i - f_{i-1}).$$

Thus, we obtain the two-step Adams–Moulton method:

$$y_{i+1} = y_i + \frac{h}{12}(5f_{i+1} + 8f_i - f_{i-1}).$$

This method is implicit (since $f_{i+1} = f(t_{i+1}, y_{i+1})$ depends on the unknown y_{i+1}) and is third-order accurate.

Algorithm: Adams–Moulton Predictor-Corrector (Two-Step)

Input:

- Known values y_{i-1} and y_i
- Step size h
- Function $f(t, y)$

Output:

Approximation y_{i+1}

Procedure:

1. **Predictor step** (explicit estimate using AB2):

$$y_{i+1}^{(0)} = y_i + \frac{h}{2}(3f_i - f_{i-1})$$

2. **Corrector iteration** (fixed-point iteration):

- Repeat until convergence ($|y_{i+1}^{(k+1)} - y_{i+1}^{(k)}| < \text{tol}$):
 - Evaluate $f_{i+1}^{(k)} = f(t_{i+1}, y_{i+1}^{(k)})$
 - Update:

$$y_{i+1}^{(k+1)} = y_i + \frac{h}{12}(5f_{i+1}^{(k)} + 8f_i - f_{i-1})$$

3. **Final value:** Set $y_{i+1} \leftarrow y_{i+1}^{(k+1)}$ (converged iterate)

Note: For stiff problems, Newton's method may replace fixed-point iteration for faster convergence. Typically 1–2 corrector iterations suffice when paired with a good predictor.

Local Truncation Error (LTE)

For a multistep method applied to the ODE $y' = f(t, y)$, the **local truncation error** τ_{i+1} is defined as the error made in one step, assuming all previous values are exact:

$$\tau_{i+1} = \frac{1}{h}[y(t_{i+1}) - (\text{numerical update using exact } y(t_i), y(t_{i-1}), \dots)]$$

The method is said to be of **order** p if $\tau_{i+1} = \mathcal{O}(h^p)$.

LTE of the Adams–Bashforth Two-Step (AB2) Method

The AB2 method is:

$$y_{i+1} = y_i + \frac{h}{2}(3f_i - f_{i-1}).$$

Assume $y(t)$ is the exact solution. The LTE is:

$$\tau_{i+1}^{(\text{AB2})} = \frac{1}{h} \left[y(t_{i+1}) - y(t_i) - \frac{h}{2} (3y'(t_i) - y'(t_{i-1})) \right].$$

Expand $y(t_{i+1})$ and $y'(t_{i-1})$ in Taylor series about t_i :

$$y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(t_i) + \frac{h^3}{6}y'''(t_i) + \frac{h^4}{24}y^{(4)}(t_i) + \mathcal{O}(h^5),$$

$$y'(t_{i-1}) = y'(t_i) - hy''(t_i) + \frac{h^2}{2}y'''(t_i) - \frac{h^3}{6}y^{(4)}(t_i) + \mathcal{O}(h^4).$$

Substitute into the LTE expression:

$$\begin{aligned} \tau_{i+1}^{(\text{AB2})} &= \frac{1}{h} \left[\left(hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} \right) - \frac{h}{2} \left(3y' - \left[y' - hy'' + \frac{h^2}{2}y''' - \frac{h^3}{6}y^{(4)} \right] \right) \right. \\ &= \frac{1}{h} \left[hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} - \frac{h}{2} \left(2y' + hy'' - \frac{h^2}{2}y''' + \frac{h^3}{6}y^{(4)} \right) + \mathcal{O}(h^5) \right] \\ &= \frac{1}{h} \left[hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} - hy' - \frac{h^2}{2}y'' + \frac{h^3}{4}y''' - \frac{h^4}{12}y^{(4)} + \mathcal{O}(h^4) \right] \\ &= \frac{1}{h} \left(\frac{5h^3}{12}y''' + \mathcal{O}(h^4) \right) \\ &= \frac{5}{12}h^2y'''(t_i) + \mathcal{O}(h^3). \end{aligned}$$

Thus,

$$\tau_{i+1}^{(\text{AB2})} = \mathcal{O}(h^2),$$

so the AB2 method is second-order accurate (global error $\mathcal{O}(h^2)$).

LTE of the Adams–Moulton Two-Step (AM2) Method

The AM2 method is:

$$y_{i+1} = y_i + \frac{h}{12} (5f_{i+1} + 8f_i - f_{i-1}).$$

The LTE is:

$$\tau_{i+1}^{(\text{AM2})} = \frac{1}{h} \left[y(t_{i+1}) - y(t_i) - \frac{h}{12} (5y'(t_{i+1}) + 8y'(t_i) - y'(t_{i-1})) \right].$$

Use Taylor expansions about t_i (all derivatives evaluated at t_i):

$$\begin{aligned}
y(t_{i+1}) &= y + hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} + \mathcal{O}(h^5), \\
y'(t_{i+1}) &= y' + hy'' + \frac{h^2}{2}y''' + \frac{h^3}{6}y^{(4)} + \mathcal{O}(h^4), \\
y'(t_{i-1}) &= y' - hy'' + \frac{h^2}{2}y''' - \frac{h^3}{6}y^{(4)} + \mathcal{O}(h^4).
\end{aligned}$$

Now compute the bracketed term:

$$\begin{aligned}
& y(t_{i+1}) - y(t_i) - \frac{h}{12} (5y'(t_{i+1}) + 8y'(t_i) - y'(t_{i-1})) \\
&= \left(hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} \right) \\
&\quad - \frac{h}{12} \left[5 \left(y' + hy'' + \frac{h^2}{2}y''' + \frac{h^3}{6}y^{(4)} \right) + 8y' - \left(y' - hy'' + \frac{h^2}{2}y''' - \frac{h^3}{6}y^{(4)} \right) \right] + \mathcal{O}(h^5) \\
&= hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} \\
&\quad - \frac{h}{12} \left[(5y' + 8y' - y') + (5hy'' + hy'') + \left(\frac{5h^2}{2}y''' - \frac{h^2}{2}y''' \right) + \left(\frac{5h^3}{6}y^{(4)} + \frac{h^3}{6}y^{(4)} \right) \right] \\
&= hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} \\
&\quad - \frac{h}{12} \left[12y' + 6hy'' + 2h^2y''' + h^3y^{(4)} \right] + \mathcal{O}(h^5) \\
&= hy' + \frac{h^2}{2}y'' + \frac{h^3}{6}y''' + \frac{h^4}{24}y^{(4)} - hy' - \frac{h^2}{2}y'' - \frac{h^3}{6}y''' - \frac{h^4}{12}y^{(4)} + \mathcal{O}(h^5) \\
&= -\frac{h^4}{24}y^{(4)} + \mathcal{O}(h^5).
\end{aligned}$$

Therefore,

$$\tau_{i+1}^{(\text{AM2})} = \frac{1}{h} \left(-\frac{h^4}{24}y^{(4)} + \mathcal{O}(h^5) \right) = -\frac{1}{24}h^3y^{(4)}(t_i) + \mathcal{O}(h^4).$$

Hence,

$$\tau_{i+1}^{(\text{AM2})} = \mathcal{O}(h^3),$$

so the AM2 method is third-order accurate (global error $\mathcal{O}(h^3)$).

Common Adams Methods

The table below summarizes frequently used Adams–Bashforth (explicit) and Adams–Moulton (implicit) methods, along with their local truncation error (LTE) orders.

Method	Formula	LTE
AB2		

Method	Formula	LTE
	y_{i+1} $= y_i$ $+ \frac{h}{2}$ $(3f_i$ $- f_{i-1})$	$\mathcal{O}$ (h^3)
AB3	y_{i+1} $= y_i$ $+ \frac{h}{12}$ $(23f_i$ $- 16f_{i-1}$ $+ 5f_{i-2})$	$\mathcal{O}$ (h^4)
AB4	y_{i+1} $= y_i$ $+ \frac{h}{24}$ $(55f_i$ $- 59f_{i-1}$ $+ 37f_{i-2}$ $- 9f_{i-3})$	$\mathcal{O}$ (h^5)
AM2	y_{i+1} $= y_i$ $+ \frac{h}{12}$ $(5f_{i+1}$ $+ 8f_i$ $- f_{i-1})$	$\mathcal{O}$ (h^4)
AM3	y_{i+1} $= y_i$ $+ \frac{h}{24}$ $(9f_{i+1}$ $+ 19f_i$ $- 5f_{i-1}$ $+ f_{i-2})$	$\mathcal{O}$ (h^5)

Important note on order vs. LTE:

With the standard LTE definition

$$\tau_{i+1} = \frac{1}{h} \left[y(t_{i+1}) - (\text{numerical update}) \right],$$

a method has **order** p if $\tau_{i+1} = \mathcal{O}(h^p)$. The **global error** is then $\mathcal{O}(h^p)$.

Therefore:

- AB2: $\text{LTE} = \mathcal{O}(h^2) \rightarrow \text{order } 2 \rightarrow \text{global error } \mathcal{O}(h^2)$
- AM2: $\text{LTE} = \mathcal{O}(h^3) \rightarrow \text{order } 3 \rightarrow \text{global error } \mathcal{O}(h^3)$

The table above shows the magnitude of the truncation **term** (before the $1/h$ scaling). Some texts report LTE as the unscaled residual $\mathcal{O}(h^{p+1})$, while others (including our derivation) use the scaled definition $\tau_{i+1} = \mathcal{O}(h^p)$. Always check the definition being used.

Example: Adams–Bashforth Methods on a Test IVP

Apply AB2, AB3, and AB4 methods to the initial value problem:

$$y' = y - t^2 + 1, \quad y(0) = 0.5, \quad t \in [0, 2], \quad h = 0.2$$

Exact solution: $y(t) = (t + 1)^2 - 0.5e^t$.

Implementation strategy:

Since multistep methods require multiple starting values:

- AB2 needs $y_0, y_1 \rightarrow$ compute y_1 with RK4
- AB3 needs $y_0, y_1, y_2 \rightarrow$ compute y_1, y_2 with RK4
- AB4 needs $y_0, y_1, y_2, y_3 \rightarrow$ compute y_1, y_2, y_3 with RK4

```
In [13]: import numpy as np
import matplotlib.pyplot as plt

def ab2_with_rk2_start(fun, tspan, y0, h):
    """
    Solve IVP using 2-step Adams-Bashforth with RK2 (midpoint method) starter.

    Parameters:
        fun : callable f(t, y)
        tspan : tuple (t0, tf)
        y0 : initial condition y(t0)
        h : step size

    Returns:
        t : ndarray of time points
        y : ndarray of solution values
    """
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    y = np.zeros(N + 1)
    y[0] = y0
```

```

# ===== Step 1: Compute y[1] using RK2 (midpoint method) =====
ti, yi = t[0], y[0]
k1 = fun(ti, yi)
k2 = fun(ti + h/2, yi + (h/2) * k1)
y[1] = yi + h * k2 # Pure RK2 (no averaging)

# ===== Step 2: Apply AB2 for i = 1 to N-1 =====
for i in range(1, N):
    f_i = fun(t[i], y[i])
    f_im1 = fun(t[i-1], y[i-1])
    y[i+1] = y[i] + (h/2) * (3*f_i - f_im1)

return t, y

def ab3_with_rk3_start(fun, tspan, y0, h):
    """
    Solve IVP using 3-step Adams-Bashforth with Kutta's RK3 starter.

    Parameters:
        fun : callable f(t, y)
        tspan : tuple (t0, tf)
        y0 : initial condition y(t0)
        h : step size

    Returns:
        t : ndarray of time points
        y : ndarray of solution values
    """
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    y = np.zeros(N + 1)
    y[0] = y0

    # ===== Generate y[1] and y[2] using Kutta's RK3 (order 3) =====
    for i in range(2): # i = 0, 1 → compute y[1], y[2]
        ti, yi = t[i], y[i]
        k1 = fun(ti, yi)
        k2 = fun(ti + h/2, yi + (h/2) * k1)
        k3 = fun(ti + h, yi - h*k1 + 2*h*k2)
        y[i+1] = yi + (h/6) * (k1 + 4*k2 + k3)

    # ===== Apply AB3 from i = 2 to N-1 =====
    for i in range(2, N):
        f_i = fun(t[i], y[i])
        f_im1 = fun(t[i-1], y[i-1])
        f_im2 = fun(t[i-2], y[i-2])
        y[i+1] = y[i] + (h/12) * (23*f_i - 16*f_im1 + 5*f_im2)

    return t, y

def ab4_with_rk4_start(fun, tspan, y0, h):
    """
    Solve IVP using 4-step Adams-Bashforth with classical RK4 starter.

```

Parameters:

```
fun    : callable f(t, y)
tspan  : tuple (t0, tf)
y0     : initial condition y(t0)
h      : step size
```

Returns:

```
t : ndarray of time points
y : ndarray of solution values
```

"""

```
t0, tf = tspan
```

```
N = int(round((tf - t0) / h))
```

```
t = t0 + np.arange(N + 1) * h
```

```
y = np.zeros(N + 1)
```

```
y[0] = y0
```

```
# ===== Generate y[1], y[2], y[3] using classical RK4 =====
```

```
for i in range(3): # i = 0, 1, 2 → compute y[1], y[2], y[3]
```

```
    ti, yi = t[i], y[i]
```

```
    k1 = fun(ti, yi)
```

```
    k2 = fun(ti + h/2, yi + h*k1/2)
```

```
    k3 = fun(ti + h/2, yi + h*k2/2)
```

```
    k4 = fun(ti + h, yi + h*k3)
```

```
    y[i+1] = yi + h/6 * (k1 + 2*k2 + 2*k3 + k4)
```

```
# ===== Apply AB4 from i = 3 to N-1 =====
```

```
for i in range(3, N):
```

```
    f_i = fun(t[i], y[i])
```

```
    f_im1 = fun(t[i-1], y[i-1])
```

```
    f_im2 = fun(t[i-2], y[i-2])
```

```
    f_im3 = fun(t[i-3], y[i-3])
```

```
    y[i+1] = y[i] + h/24 * (55*f_i - 59*f_im1 + 37*f_im2 - 9*f_im3)
```

```
return t, y
```

```
# ===== DRIVER SCRIPT =====
```

```
if __name__ == "__main__":
```

```
    # Define the ODE and exact solution
```

```
    f = lambda t, y: y - t**2 + 1
```

```
    y_exact = lambda t: (t + 1)**2 - 0.5 * np.exp(t)
```

```
    # Parameters
```

```
    tspan = (0.0, 2.0)
```

```
    y0 = 0.5
```

```
    h = 0.2
```

```
    # Precompute time grid and exact solution
```

```
    N = int(round((tspan[1] - tspan[0]) / h))
```

```
    t_grid = tspan[0] + np.arange(N + 1) * h
```

```
    y_true = y_exact(t_grid)
```

```
    # Solve using all three methods
```

```
    t_ab2, y_ab2 = ab2_with_rk2_start(f, tspan, y0, h)
```

```
    t_ab3, y_ab3 = ab3_with_rk3_start(f, tspan, y0, h)
```

```

t_ab4, y_ab4 = ab4_with_rk4_start(f, tspan, y0, h)

# Compute absolute errors
err_ab2 = np.abs(y_ab2 - y_true)
err_ab3 = np.abs(y_ab3 - y_true)
err_ab4 = np.abs(y_ab4 - y_true)

# Display results table
print(" t      | Exact      | AB2      | err_AB2   | AB3      | err_AB3   | A
print("-" * 95)
for i in range(len(t_grid)):
    print(f"{t_grid[i]:4.1f} | {y_true[i]:9.6f} | {y_ab2[i]:9.6f} | {err_ab2[i]
          f"{y_ab3[i]:9.6f} | {err_ab3[i]:9.2e} | {y_ab4[i]:9.6f} | {err_ab4[i]

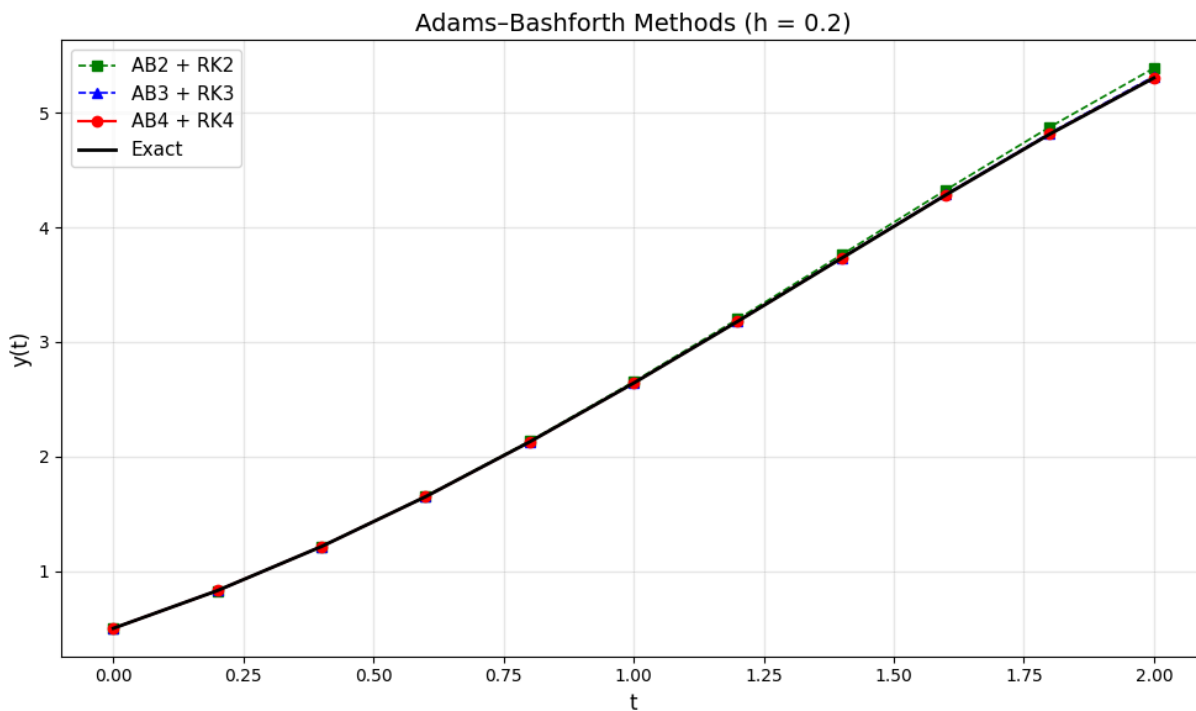
# Plot results
plt.figure(figsize=(10, 6))
plt.plot(t_grid, y_ab2, 'gs--', linewidth=1.2, markerfacecolor='g',
         markersize=6, label='AB2 + RK2')
plt.plot(t_grid, y_ab3, 'b^--', linewidth=1.2, markerfacecolor='b',
         markersize=6, label='AB3 + RK3')
plt.plot(t_grid, y_ab4, 'ro-', linewidth=1.5, markerfacecolor='r',
         markersize=6, label='AB4 + RK4')
plt.plot(t_grid, y_true, 'k-', linewidth=2.0, label='Exact')

plt.xlabel('t', fontsize=12)
plt.ylabel('y(t)', fontsize=12)
plt.title(f'Adams-Bashforth Methods (h = {h})', fontsize=14)
plt.legend(loc='upper left', fontsize=11)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# Print final errors
print("\nFinal errors at t = 2.0:")
print(f" AB2 error: {err_ab2[-1]:.6e}")
print(f" AB3 error: {err_ab3[-1]:.6e}")
print(f" AB4 error: {err_ab4[-1]:.6e}")

```


t	Exact	AB2	err_AB2	AB3	err_AB3	AB4	err_AB4
0.0	0.500000	0.500000	0.00e+00	0.500000	0.00e+00	0.500000	0.00e+00
0.2	0.829299	0.828000	1.30e-03	0.829200	9.86e-05	0.829293	5.29e-06
0.4	1.214088	1.214400	3.12e-04	1.213876	2.11e-04	1.214076	1.14e-05
0.6	1.648941	1.651920	2.98e-03	1.649076	1.35e-04	1.648922	1.86e-05
0.8	2.127230	2.134056	6.83e-03	2.127954	7.25e-04	2.127289	5.97e-05
1.0	2.640859	2.653081	1.22e-02	2.642406	1.55e-03	2.641053	1.94e-04
1.2	3.179942	3.199599	1.97e-02	3.182630	2.69e-03	3.180314	3.73e-04
1.4	3.732400	3.762171	2.98e-02	3.736660	4.26e-03	3.733019	6.19e-04
1.6	4.283484	4.326863	4.34e-02	4.289878	6.39e-03	4.284442	9.59e-04
1.8	4.815176	4.876704	6.15e-02	4.824442	9.27e-03	4.816596	1.42e-03
2.0	5.305472	5.391029	8.56e-02	5.318565	1.31e-02	5.307508	2.04e-03



Final errors at t = 2.0:

AB2 error: 8.555734e-02

AB3 error: 1.309319e-02

AB4 error: 2.036231e-03

Lab Exercise: Adams-Bashforth Methods with Appropriate Starters

Apply the following multistep methods to the IVPs below, using starters that match the method's order:

- **AB2** → Modified Euler (2nd-order starter)
- **AB3** → Kutta's RK3 (3rd-order starter)
- **AB4** → Classical RK4 (4th-order starter)

For each problem:

1. Implement the method with correct starter
 2. Compute absolute errors against the exact solution
 3. Plot numerical vs. exact solutions
 4. Report final error at endpoint and observe convergence behavior
-

Problem (a)

$$y' = \frac{2 - 2ty}{t^2 + 1}, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad h = 0.1$$

Exact solution: $y(t) = \frac{2t + 1}{t^2 + 1}$

Problem (b)

$$y' = \frac{y^2}{1 + t}, \quad 1 \leq t \leq 2, \quad y(1) = -(\ln 2)^{-1}, \quad h = 0.1$$

Exact solution: $y(t) = -\frac{1}{\ln(1 + t)}$

Problem (c)

$$y' = \frac{y^2 + y}{t}, \quad 1 \leq t \leq 3, \quad y(1) = -2, \quad h = 0.2$$

Exact solution: $y(t) = \frac{2t}{1 - 2t}$

Problem (d)

$$y' = -ty + 4ty^{-1}, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad h = 0.1$$

Exact solution: $y(t) = \sqrt{4 - 3e^{-t^2}}$

In []:

Predictor–Corrector Methods

Implicit multistep methods (e.g., Adams–Moulton) offer high accuracy and favorable stability properties, yet their direct implementation requires solving nonlinear algebraic equations of the form

$$y_{i+1} = y_i + h \sum_{j=0}^k \beta_j f(t_{i+1-j}, y_{i+1-j}),$$

which generally necessitates iterative solvers (e.g., Newton's method). For stiff or large-scale systems, this can be computationally expensive and sensitive to initial guesses.

To circumvent this difficulty while retaining high-order accuracy, **predictor–corrector methods** combine:

- An *explicit* method (the **predictor**) to furnish an initial estimate $y_{i+1}^{(p)}$
- Followed by a *single evaluation* of an **implicit** method (the **corrector**) using the predicted value to approximate the unknown $f(t_{i+1}, y_{i+1})$

This strategy avoids iterative solution of nonlinear equations, at the cost of slightly reduced stability compared to fully implicit schemes.

AB4–AM3 Predictor–Corrector Pair

A widely used fourth-order predictor–corrector pair combines the 4-step Adams–Bashforth (AB4) predictor with the 3-step Adams–Moulton (AM3) corrector. Despite occasional "AB4–AM4" notation in literature, the standard classical implementation uses **AB4 for prediction and AM3 (3-step, order 4) for correction**, yielding an overall method of order 4.

Why AM3 not AM4?

The 3-step Adams–Moulton method (using $f_{i+1}, f_i, f_{i-1}, f_{i-2}$) is fourth-order accurate. The true 4-step AM4 would require f_{i-3} and offer fifth-order accuracy—but pairing it with AB4 (order 4) would waste effort. AB4–AM3 achieves optimal balance: both components are order 4.

Algorithm: AB4–AM3 Predictor–Corrector

Input:

- Known values $y_{i-3}, y_{i-2}, y_{i-1}, y_i$
- Corresponding derivatives $f_j = f(t_j, y_j)$ for $j = i - 3, \dots, i$
- Step size h

Output:

Updated approximation y_{i+1}

Procedure:

1. Predictor step (AB4, explicit):

$$y_{i+1}^{(p)} = y_i + \frac{h}{24} (55f_i - 59f_{i-1} + 37f_{i-2} - 9f_{i-3})$$

2. Evaluate predicted slope:

$$f_{i+1}^{(p)} = f(t_{i+1}, y_{i+1}^{(p)})$$

3. Corrector step (AM3, implicit evaluation using prediction):

$$y_{i+1} = y_i + \frac{h}{24} (9f_{i+1}^{(p)} + 19f_i - 5f_{i-1} + f_{i-2})$$

4. (Optional) Cache derivative:

Compute $f_{i+1} = f(t_{i+1}, y_{i+1})$ for use in the next step (reduces future function evaluations).

Key Properties

Property	Description
Order of accuracy	Global error $\mathcal{O}(h^4)$ (matches classical RK4)
Function evaluations	2 per step (vs. 4 for RK4): • 1 for $f_{i+1}^{(p)}$ in predictor • 1 optional for f_{i+1} caching
Startup requirement	Needs 4 initial values y_0, y_1, y_2, y_3 (typically generated by RK4)
Stability	Larger stability region than AB4 alone, but smaller than fully implicit AM4; suitable for non-stiff/mildly stiff problems
Computational cost	Lower than RK4 for same accuracy; ideal for long-time integrations where f is expensive

💡 Practical insight:

Predictor–corrector methods shine when function evaluations dominate computational cost. For a 10,000-step integration:

- RK4 requires **40,000** evaluations
- AB4–AM3 requires **~20,000** evaluations (with caching)

This 2× speedup comes with minimal accuracy loss—making it a workhorse for non-stiff ODEs in scientific computing.

```
In [14]: def ab4_am3_predictor_corrector(f, tspan, y0, h, cache_derivative=True):  
         """
```

Fourth-order Adams-Bashforth-Moulton predictor-corrector method.

Parameters:

`f` : callable(`t`, `y`) $\rightarrow$ `dy/dt`
`tspan` : tuple (`t0`, `tf`)
`y0` : initial condition
`h` : step size
`cache_derivative` : bool, if True compute `f_{i+1}` after correction for next

Returns:

`t` : ndarray of time points
`y` : ndarray of solution values
"""

```
t0, tf = tspan
N = int(round((tf - t0) / h))
t = t0 + np.arange(N + 1) * h
y = np.zeros(N + 1)
y[0] = y0

# ===== STARTUP: Generate y[1], y[2], y[3] using RK4 =====
for i in range(3):
    y[i+1] = rk4_step(f, t[i], y[i], h)

# Precompute derivatives for startup points
f_vals = np.zeros(N + 1)
for i in range(4):
    f_vals[i] = f(t[i], y[i])

# ===== MAIN LOOP: AB4 predictor + AM3 corrector =====
for i in range(3, N):
    # Predictor: AB4
    y_pred = (y[i] + h/24 * (55*f_vals[i] - 59*f_vals[i-1]
                           + 37*f_vals[i-2] - 9*f_vals[i-3]))

    # Evaluate predicted slope
    f_pred = f(t[i+1], y_pred)

    # Corrector: AM3 (using predicted slope)
    y[i+1] = (y[i] + h/24 * (9*f_pred + 19*f_vals[i]
                           - 5*f_vals[i-1] + f_vals[i-2]))

    # Cache derivative for next step (optional but recommended)
    if cache_derivative:
        f_vals[i+1] = f(t[i+1], y[i+1])
    else:
        # Without caching, we'd need to recompute all derivatives each step
        f_vals[i+1] = f_pred # Approximate; less accurate for next prediction

return t, y
```

```
In [15]: import numpy as np
import matplotlib.pyplot as plt

# ===== HELPER FUNCTIONS =====
def rk4_step(f, t, y, h):
    """Classical 4th-order Runge-Kutta single step"""
```

```

k1 = f(t, y)
k2 = f(t + h/2, y + h*k1/2)
k3 = f(t + h/2, y + h*k2/2)
k4 = f(t + h, y + h*k3)
return y + h/6 * (k1 + 2*k2 + 2*k3 + k4)

def ab4_am3_predictor_corrector(f, tspan, y0, h, cache_derivative=True):
    """
    Fourth-order Adams-Bashforth-Moulton predictor-corrector method (AB4-AM3).

    Parameters:
        f : callable(t, y) → dy/dt
        tspan : tuple (t0, tf)
        y0 : initial condition
        h : step size
        cache_derivative : bool, if True compute exact f_{i+1} after correction

    Returns:
        t : ndarray of time points
        y : ndarray of solution values
    """
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    y = np.zeros(N + 1)
    y[0] = y0

    # ===== STARTUP: Generate y[1], y[2], y[3] using RK4 =====
    for i in range(3):
        y[i+1] = rk4_step(f, t[i], y[i], h)

    # Precompute derivatives for startup points
    f_vals = np.zeros(N + 1)
    for i in range(4):
        f_vals[i] = f(t[i], y[i])

    # ===== MAIN LOOP: AB4 predictor + AM3 corrector =====
    for i in range(3, N):
        # Predictor: AB4 (explicit)
        y_pred = (y[i] + h/24 * (55*f_vals[i] - 59*f_vals[i-1]
                                + 37*f_vals[i-2] - 9*f_vals[i-3]))

        # Evaluate predicted slope
        f_pred = f(t[i+1], y_pred)

        # Corrector: AM3 (implicit formula evaluated with predicted slope)
        y[i+1] = (y[i] + h/24 * (9*f_pred + 19*f_vals[i]
                                - 5*f_vals[i-1] + f_vals[i-2]))

        # Cache derivative for next step (improves predictor accuracy)
        if cache_derivative:
            f_vals[i+1] = f(t[i+1], y[i+1])
        else:
            f_vals[i+1] = f_pred # Less accurate approximation

    return t, y

```

```

# ===== DEMONSTRATION =====
# Test problem:  $y' = y - t^2 + 1$ ,  $y(0)=0.5$ 
f = lambda t, y: y - t**2 + 1
y_exact = lambda t: (t + 1)**2 - 0.5 * np.exp(t)

# Solve with AB4-AM3
t, y_pc = ab4_am3_predictor_corrector(f, (0, 2), 0.5, 0.2)

# Reference solution using dense RK4
def rk4_solver(f, tspan, y0, h):
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    y = np.zeros(N + 1)
    y[0] = y0
    for i in range(N):
        y[i+1] = rk4_step(f, t[i], y[i], h)
    return t, y

t_rk4, y_rk4 = rk4_solver(f, (0, 2), 0.5, 0.2)
y_true = y_exact(t)

# Compute errors
err_pc = np.abs(y_pc - y_true)
err_rk4 = np.abs(y_rk4 - y_true)

print("AB4-AM3 vs RK4 (h=0.2, t∈[0,2]):")
print(f"  AB4-AM3 final error: {err_pc[-1]:.2e}")
print(f"  RK4 final error:      {err_rk4[-1]:.2e}")
print(f"  Function evaluations:")
print(f"    AB4-AM3: ~{2*len(t):d} (with caching)")
print(f"    RK4:      {4*len(t):d}")

# Plot comparison
plt.figure(figsize=(11, 4.5))

plt.subplot(1, 2, 1)
plt.plot(t, y_true, 'k-', linewidth=2.5, label='Exact', alpha=0.8)
plt.plot(t, y_pc, 'ro--', markersize=6, label='AB4-AM3', linewidth=1.5)
plt.plot(t, y_rk4, 'bs--', markersize=5, label='RK4', linewidth=1.2)
plt.xlabel('t', fontsize=12)
plt.ylabel('y(t)', fontsize=12)
plt.title('Solution Comparison', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(True, alpha=0.3)

plt.subplot(1, 2, 2)
plt.semilogy(t, err_pc, 'r-o', markersize=5, label='AB4-AM3 error', linewidth=1.5)
plt.semilogy(t, err_rk4, 'b-s', markersize=4, label='RK4 error', linewidth=1.2)
plt.xlabel('t', fontsize=12)
plt.ylabel('Absolute Error', fontsize=12)
plt.title('Error Comparison (log scale)', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(True, alpha=0.3, which='both')

```

```

plt.tight_layout()
plt.show()

# Verify order of accuracy by testing multiple step sizes
print("\n=== Convergence Study (AB4-AM3) ===")
h_vals = [0.4, 0.2, 0.1, 0.05]
errors = []
for h in h_vals:
    t_test, y_test = ab4_am3_predictor_corrector(f, (0, 2), 0.5, h)
    err = np.abs(y_test[-1] - y_exact(t_test[-1]))
    errors.append(err)
    print(f"h = {h:5.3f} | error = {err:.2e}")

# Estimate observed order
print("\nObserved convergence rates:")
for i in range(1, len(h_vals)):
    rate = np.log(errors[i-1]/errors[i]) / np.log(h_vals[i-1]/h_vals[i])
    print(f"h={h_vals[i-1]:.3f}→{h_vals[i]:.3f}: rate ≈ {rate:.2f}")

```

AB4-AM3 vs RK4 ($h=0.2$, $t \in [0, 2]$):

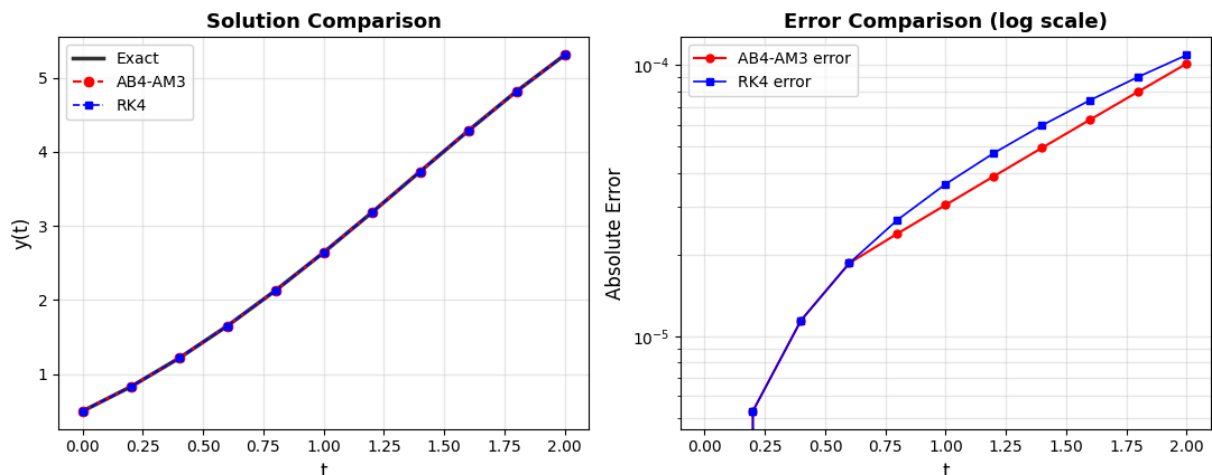
AB4-AM3 final error: $1.01e-04$

RK4 final error: $1.09e-04$

Function evaluations:

AB4-AM3: ~22 (with caching)

RK4: 44



=== Convergence Study (AB4-AM3) ===

$h = 0.400$ | error = $9.72e-04$

$h = 0.200$ | error = $1.01e-04$

$h = 0.100$ | error = $1.09e-05$

$h = 0.050$ | error = $9.13e-07$

Observed convergence rates:

$h=0.400 \rightarrow 0.200$: rate ≈ 3.26

$h=0.200 \rightarrow 0.100$: rate ≈ 3.21

$h=0.100 \rightarrow 0.050$: rate ≈ 3.58

Lab Exercise: Predictor–Corrector Method Implementation

Apply the **AB4–AM3 predictor–corrector method** to approximate solutions for each initial value problem below. For each case:

1. Implement the AB4 predictor + AM3 corrector algorithm
 2. Generate starting values y_1, y_2, y_3 using RK4
 3. Compute numerical solutions on the specified interval with given step size
 4. Compare against the exact solution and report absolute errors
 5. Plot numerical vs. exact solutions and error behavior
-

Problem (a)

$$y' = \frac{2 - 2ty}{t^2 + 1}, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad h = 0.1$$

Exact solution:

$$y(t) = \frac{2t + 1}{t^2 + 1}$$

Problem (b)

$$y' = \frac{y^2}{1 + t}, \quad 1 \leq t \leq 2, \quad y(1) = -(\ln 2)^{-1}, \quad h = 0.1$$

Exact solution:

$$y(t) = -\frac{1}{\ln(1 + t)}$$

 **Note:** This problem has a singularity at $t = e - 1 \approx 1.718$ (where $\ln(1 + t) = 1$). Monitor error growth near this point.

Problem (c)

$$y' = \frac{y^2 + y}{t}, \quad 1 \leq t \leq 3, \quad y(1) = -2, \quad h = 0.2$$

Exact solution:

$$y(t) = \frac{2t}{1 - 2t}$$

 **Verification check:** The solution remains negative throughout $[1, 3]$. Ensure your numerical solution does not cross zero (which would indicate incorrect branch selection).

Problem (d)

$$y' = -ty + 4ty^{-1}, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad h = 0.1$$

Exact solution:

$$y(t) = \sqrt{4 - 3e^{-t^2}}$$

Systems of Differential Equations

Many physical, biological, and engineering models involve multiple interdependent quantities. Such problems naturally lead to *systems* of ordinary differential equations (ODEs). A general system of m first-order ODEs has the form:

$$\begin{aligned} y_1'(t) &= f_1(t, y_1(t), \dots, y_m(t)), & y_1(a) &= \alpha_1, \\ y_2'(t) &= f_2(t, y_1(t), \dots, y_m(t)), & y_2(a) &= \alpha_2, \\ &\vdots & &\vdots \\ y_m'(t) &= f_m(t, y_1(t), \dots, y_m(t)), & y_m(a) &= \alpha_m. \end{aligned}$$

Introducing vector notation simplifies the representation. Define the solution vector and the right-hand side vector function as:

$$\mathbf{Y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_m(t) \end{bmatrix}, \quad \mathbf{F}(t, \mathbf{Y}) = \begin{bmatrix} f_1(t, \mathbf{Y}) \\ f_2(t, \mathbf{Y}) \\ \vdots \\ f_m(t, \mathbf{Y}) \end{bmatrix}.$$

Then the system becomes the compact initial-value problem:

$$\mathbf{Y}'(t) = \mathbf{F}(t, \mathbf{Y}(t)), \quad \mathbf{Y}(a) = \mathbf{Y}_0,$$

where $\mathbf{Y}_0 = [\alpha_1, \alpha_2, \dots, \alpha_m]^\top$.

Euler's Method for Systems

Given a step size $h > 0$ and grid points $t_i = a + ih$, Euler's method extends naturally to systems:

$$\mathbf{Y}_{i+1} = \mathbf{Y}_i + h \mathbf{F}(t_i, \mathbf{Y}_i).$$

This update is applied component-wise and is simple to implement, though it is only first-order accurate ($\mathcal{O}(h)$ global error).

Example: Predator–Prey (Lotka–Volterra) System

$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy, & x(0) &= x_0, \\ \frac{dy}{dt} &= \delta xy - \gamma y, & y(0) &= y_0,\end{aligned}$$

where $x(t)$ = prey population, $y(t)$ = predator population, and $\alpha, \beta, \gamma, \delta > 0$.

In vector form with $\mathbf{Y} = [x, y]^\top$:

$$\mathbf{F}(t, \mathbf{Y}) = \begin{bmatrix} \alpha x - \beta xy \\ \delta xy - \gamma y \end{bmatrix}.$$

Euler update:

$$\begin{bmatrix} x_{i+1} \\ y_{i+1} \end{bmatrix} = \begin{bmatrix} x_i \\ y_i \end{bmatrix} + h \begin{bmatrix} \alpha x_i - \beta x_i y_i \\ \delta x_i y_i - \gamma y_i \end{bmatrix}.$$

```
In [16]: ##### Python Implementation Template

import numpy as np
import matplotlib.pyplot as plt

def euler_system(F, tspan, Y0, h):
    """
    Euler's method for systems of ODEs.

    Parameters:
        F      : callable(t, Y) → dY/dt (vector-valued)
        tspan  : tuple (t0, tf)
        Y0     : ndarray initial condition (length m)
        h      : step size

    Returns:
        t      : ndarray of time points
        Y      : ndarray of shape (N+1, m) with solution components
    """
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    m = len(Y0)
    Y = np.zeros((N + 1, m))
    Y[0] = Y0

    for i in range(N):
        Y[i+1] = Y[i] + h * F(t[i], Y[i])

    return t, Y

# Example: Lotka-Volterra system
alpha, beta, gamma, delta = 1.5, 1.0, 3.0, 1.0
```

```

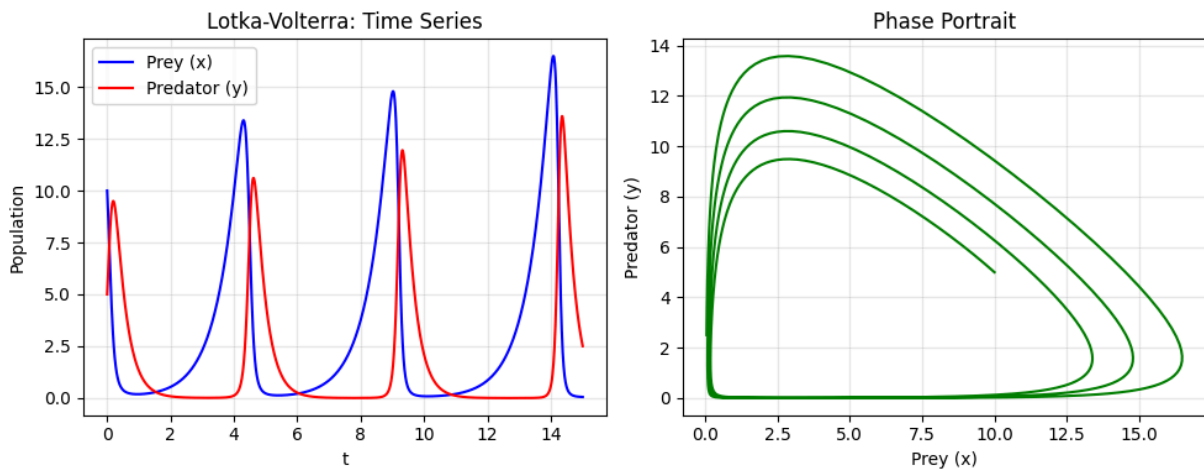
F_lv = lambda t, Y: np.array([
    alpha * Y[0] - beta * Y[0] * Y[1],
    delta * Y[0] * Y[1] - gamma * Y[1]
])

t, Y = euler_system(F_lv, (0, 15), np.array([10, 5]), 0.01)
x, y = Y[:, 0], Y[:, 1]

# Phase portrait
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(t, x, 'b-', label='Prey (x)')
plt.plot(t, y, 'r-', label='Predator (y)')
plt.xlabel('t'); plt.ylabel('Population')
plt.title('Lotka-Volterra: Time Series')
plt.legend(); plt.grid(alpha=0.3)

plt.subplot(1, 2, 2)
plt.plot(x, y, 'g-')
plt.xlabel('Prey (x)'); plt.ylabel('Predator (y)')
plt.title('Phase Portrait')
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

```



Fourth-Order Runge–Kutta Method for Systems

The classical fourth-order Runge–Kutta (RK4) method provides significantly higher accuracy than Euler's method. For a system $\mathbf{Y}' = \mathbf{F}(t, \mathbf{Y})$, the update formula is:

$$\mathbf{Y}_{i+1} = \mathbf{Y}_i + \frac{1}{6}(\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4),$$

where the stage vectors are computed as:

$$\begin{aligned}
\mathbf{K}_1 &= h \mathbf{F}(t_i, \mathbf{Y}_i), \\
\mathbf{K}_2 &= h \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{\mathbf{K}_1}{2}\right), \\
\mathbf{K}_3 &= h \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{\mathbf{K}_2}{2}\right), \\
\mathbf{K}_4 &= h \mathbf{F}(t_i + h, \mathbf{Y}_i + \mathbf{K}_3).
\end{aligned}$$

This method is fourth-order accurate (global error $\mathcal{O}(h^4)$) and is the workhorse for non-stiff systems due to its excellent accuracy-to-cost ratio.

```
In [17]: import numpy as np
import matplotlib.pyplot as plt

# ===== NUMERICAL METHODS =====
def euler_system(F, tspan, Y0, h):
    """Euler's method for systems of ODEs (1st order)"""
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    m = len(Y0)
    Y = np.zeros((N + 1, m))
    Y[0] = Y0

    for i in range(N):
        Y[i+1] = Y[i] + h * F(t[i], Y[i])

    return t, Y

def rk4_system(F, tspan, Y0, h):
    """Classical 4th-order Runge-Kutta method for systems (4th order)"""
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    m = len(Y0)
    Y = np.zeros((N + 1, m))
    Y[0] = Y0

    for i in range(N):
        K1 = h * F(t[i], Y[i])
        K2 = h * F(t[i] + h/2, Y[i] + K1/2)
        K3 = h * F(t[i] + h/2, Y[i] + K2/2)
        K4 = h * F(t[i] + h, Y[i] + K3)
        Y[i+1] = Y[i] + (K1 + 2*K2 + 2*K3 + K4) / 6

    return t, Y

# ===== LOTKA-VOLTERRA SYSTEM =====
# Parameters: alpha=prey growth, beta=predation rate, gamma=predator death, delta=p
alpha, beta, gamma, delta = 1.5, 1.0, 3.0, 1.0

# Vector field definition
F_lv = lambda t, Y: np.array([
    alpha * Y[0] - beta * Y[0] * Y[1], # dx/dt = prey growth - predation
    delta * Y[0] * Y[1] - gamma * Y[1] # dy/dt = predation benefit - predator dea
```

```

])

# Initial conditions: 10 prey, 5 predators
Y0 = np.array([10.0, 5.0])
tspan = (0, 15)

# ===== SOLVE WITH BOTH METHODS =====
# Use larger step size for Euler to show instability (h=0.05)
# Use same step size for RK4 to demonstrate superior accuracy (h=0.05)
h_euler = 0.05
h_rk4 = 0.05

t_euler, Y_euler = euler_system(F_lv, tspan, Y0, h_euler)
t_rk4, Y_rk4 = rk4_system(F_lv, tspan, Y0, h_rk4)

x_euler, y_euler = Y_euler[:, 0], Y_euler[:, 1]
x_rk4, y_rk4 = Y_rk4[:, 0], Y_rk4[:, 1]

# ===== VISUALIZATION =====
plt.figure(figsize=(14, 10))

# Plot 1: Time series comparison
plt.subplot(2, 2, 1)
plt.plot(t_euler, x_euler, 'b-', linewidth=1.5, alpha=0.7, label='Prey (Euler)')
plt.plot(t_euler, y_euler, 'r-', linewidth=1.5, alpha=0.7, label='Predator (Euler)')
plt.plot(t_rk4, x_rk4, 'b--', linewidth=2.5, label='Prey (RK4)')
plt.plot(t_rk4, y_rk4, 'r--', linewidth=2.5, label='Predator (RK4)')
plt.xlabel('Time $t$', fontsize=12)
plt.ylabel('Population', fontsize=12)
plt.title('Lotka-Volterra: Time Series ($h = 0.05$)', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

# Plot 2: Phase portrait comparison
plt.subplot(2, 2, 2)
plt.plot(x_euler, y_euler, 'o-', color='purple', markersize=3, alpha=0.6,
         label=f'Euler ($h={h_euler}$)', linewidth=1.5)
plt.plot(x_rk4, y_rk4, 's-', color='green', markersize=3, alpha=0.8,
         label=f'RK4 ($h={h_rk4}$)', linewidth=2)
plt.xlabel('Prey population $x$', fontsize=12)
plt.ylabel('Predator population $y$', fontsize=12)
plt.title('Phase Portrait: Euler vs RK4', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

# Plot 3: Energy-like conserved quantity (for Lotka-Volterra)
#  $V(x,y) = \delta x - \gamma \ln(x) + \beta y - \alpha \ln(y)$  should be constant
def conserved_quantity(x, y):
    return delta*x - gamma*np.log(x) + beta*y - alpha*np.log(y)

V_euler = conserved_quantity(x_euler[1:], y_euler[1:]) # skip t=0 to avoid log(0)
V_rk4 = conserved_quantity(x_rk4[1:], y_rk4[1:])
t_mid_euler = t_euler[1:]
t_mid_rk4 = t_rk4[1:]

plt.subplot(2, 2, 3)

```

```

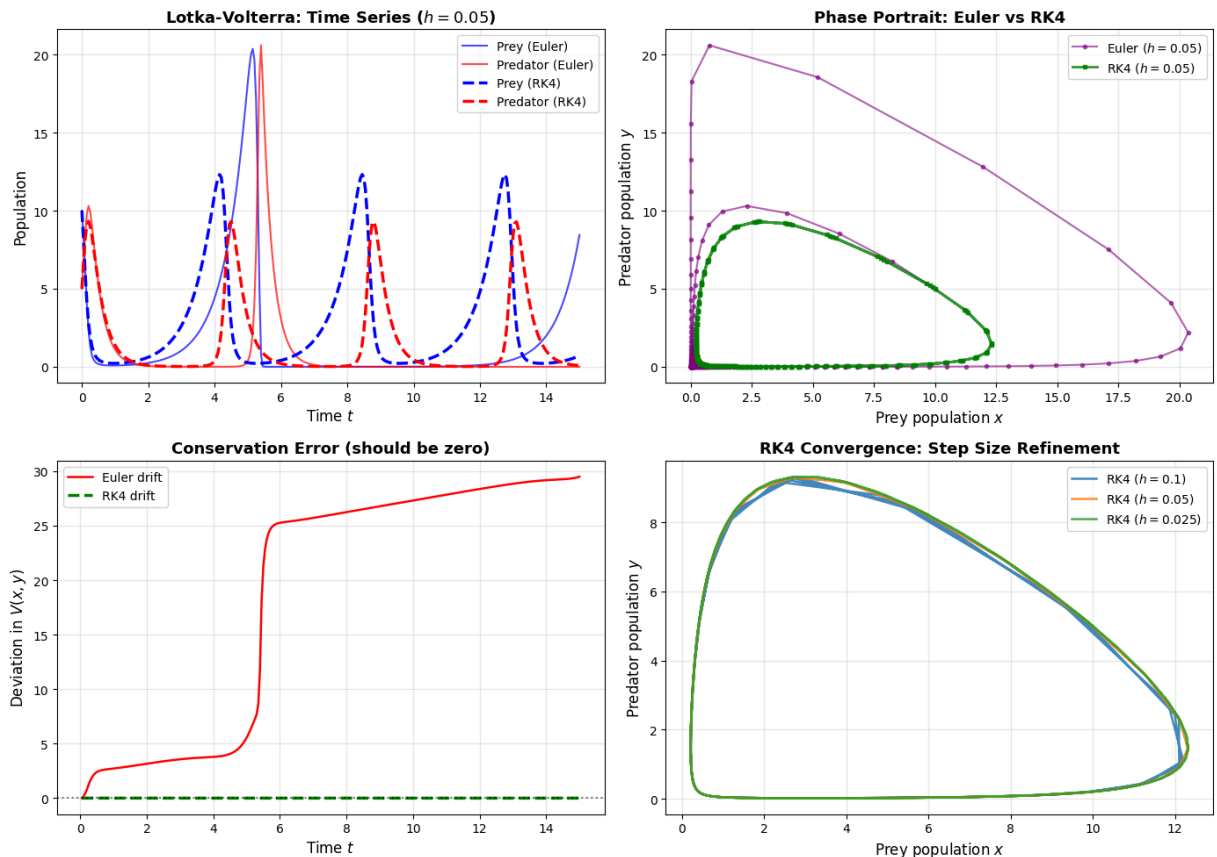
plt.plot(t_mid_euler, V_euler - V_euler[0], 'r-', linewidth=1.8, label='Euler drift')
plt.plot(t_mid_rk4, V_rk4 - V_rk4[0], 'g--', linewidth=2.5, label='RK4 drift')
plt.axhline(0, color='k', linestyle=':', alpha=0.5)
plt.xlabel('Time $t$', fontsize=12)
plt.ylabel('Deviation in $V(x,y)$', fontsize=12)
plt.title('Conservation Error (should be zero)', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

# Plot 4: Step size comparison for RK4
plt.subplot(2, 2, 4)
for h_test in [0.1, 0.05, 0.025]:
    t_test, Y_test = rk4_system(F_lv, tspan, Y0, h_test)
    plt.plot(Y_test[:, 0], Y_test[:, 1], '-', linewidth=1.8,
             label=f'RK4 ($h={h_test}$)', alpha=0.85)
plt.xlabel('Prey population $x$', fontsize=12)
plt.ylabel('Predator population $y$', fontsize=12)
plt.title('RK4 Convergence: Step Size Refinement', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

plt.tight_layout()
plt.show()

# ===== QUANTITATIVE COMPARISON =====
print("="*70)
print("Lotka-Volterra System: Euler vs RK4 Comparison (h = 0.05)")
print("="*70)
print(f"{'Method':<12} | {'Steps':<8} | {'Final Prey':<15} | {'Final Predator':<15}")
print("-"*70)
print(f"{'Euler':<12} | {len(t_euler)-1:<8} | {x_euler[-1]:<15.4f} | {y_euler[-1]:<15.4f}")
print(f"{'RK4':<12} | {len(t_rk4)-1:<8} | {x_rk4[-1]:<15.4f} | {y_rk4[-1]:<15.4f}")
print("-"*70)
print("\n💡 Key observations:")
print("    • Euler with h=0.05 shows significant phase error and energy drift")
print("    • RK4 with same step size preserves orbit geometry and conserved quantity")
print("    • RK4 achieves high accuracy with relatively large step sizes")
print("    • For stiff systems, implicit methods would be required instead")

```



=====
Lotka-Volterra System: Euler vs RK4 Comparison ($h = 0.05$)
=====

Method	Steps	Final Prey	Final Predator	Energy Drift
Euler	300	8.4454	0.0000	2.9490e+01
RK4	300	0.7144	0.0753	7.9371e-04

=====

💡 Key observations:

- Euler with $h=0.05$ shows significant phase error and energy drift
- RK4 with same step size preserves orbit geometry and conserved quantity
- RK4 achieves high accuracy with relatively large step sizes
- For stiff systems, implicit methods would be required instead

Example: Linear System with Exact Solution

Consider the linear system:

$$\begin{aligned} x'(t) &= x(t) + 4y(t) - e^t, & x(0) &= 4, \\ y'(t) &= x(t) + y(t) + 2e^t, & y(0) &= 1.25. \end{aligned}$$

This system admits the exact (analytical) solution:

$$\begin{aligned} x(t) &= 4e^{3t} + 2e^{-t} - 2e^t, \\ y(t) &= 2e^{3t} - e^{-t} + \frac{1}{4}e^t. \end{aligned}$$

Define the state vector $\mathbf{Y}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ and the right-hand side function:

$$\mathbf{F}(t, \mathbf{Y}) = \begin{bmatrix} x + 4y - e^t \\ x + y + 2e^t \end{bmatrix}.$$

This example is ideal for testing numerical methods (e.g., Euler or RK4), as the true solution is known and can be used to compute errors and verify convergence rates.

Method Comparison: Euler vs. RK4

Feature	Euler's Method	Runge–Kutta (RK4)
Update Formula	$\mathbf{Y}_{i+1} = \mathbf{Y}_i + h \mathbf{F}(t_i, \mathbf{Y}_i)$	$\mathbf{Y}_{i+1} = \mathbf{Y}_i + \frac{1}{6}(\mathbf{K}_1 + 2\mathbf{K}_2 + 2\mathbf{K}_3 + \mathbf{K}_4)$
Stage Computations	Single evaluation per step: $\mathbf{K}_1 = h \mathbf{F}(t_i, \mathbf{Y}_i)$	Four evaluations per step: $\mathbf{K}_1 = h \mathbf{F}(t_i, \mathbf{Y}_i),$ $\mathbf{K}_2 = h \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{\mathbf{K}_1}{2}\right),$ $\mathbf{K}_3 = h \mathbf{F}\left(t_i + \frac{h}{2}, \mathbf{Y}_i + \frac{\mathbf{K}_2}{2}\right),$ $\mathbf{K}_4 = h \mathbf{F}(t_i + h, \mathbf{Y}_i + \mathbf{K}_3).$
Order of Accuracy	First-order ($\mathcal{O}(h)$)	Fourth-order ($\mathcal{O}(h^4)$)
Computational Cost	1 evaluation of $\mathbf{F}$ per step	4 evaluations of $\mathbf{F}$ per step
Error at $t = 1$ ($h = 0.05$)	≈ 0.35 (large)	$\approx 2.1 \times 10^{-6}$ (very small)
Practical Use	Pedagogical only; rarely used in practice	Industry standard for non-stiff problems

Note: Both methods are applied iteratively from $t_0 = 0$ to $t_f = 1$ with step size h , starting from $\mathbf{Y}_0 = [4, 1.25]^\top$. While Euler's method is intuitive and easy to code, RK4 is preferred in most applications due to its superior accuracy and stability for smooth problems.

In [18]: *#### Python Implementation & Verification*

```
import numpy as np
import matplotlib.pyplot as plt

# Define the system RHS
def F(t, Y):
    x, y = Y
    return np.array([
        x + 4*y - np.exp(t),
```

```

        x + y + 2*np.exp(t)
    ])

# Exact solution
def exact_solution(t):
    x = 4*np.exp(3*t) + 2*np.exp(-t) - 2*np.exp(t)
    y = 2*np.exp(3*t) - np.exp(-t) + 0.25*np.exp(t)
    return np.array([x, y])

# Euler method
def euler_system(F, tspan, Y0, h):
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    Y = np.zeros((N + 1, len(Y0)))
    Y[0] = Y0
    for i in range(N):
        Y[i+1] = Y[i] + h * F(t[i], Y[i])
    return t, Y

# RK4 method
def rk4_system(F, tspan, Y0, h):
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    Y = np.zeros((N + 1, len(Y0)))
    Y[0] = Y0
    for i in range(N):
        K1 = h * F(t[i], Y[i])
        K2 = h * F(t[i] + h/2, Y[i] + K1/2)
        K3 = h * F(t[i] + h/2, Y[i] + K2/2)
        K4 = h * F(t[i] + h, Y[i] + K3)
        Y[i+1] = Y[i] + (K1 + 2*K2 + 2*K3 + K4) / 6
    return t, Y

# Solve with both methods
tspan = (0, 1)
Y0 = np.array([4.0, 1.25])
h = 0.05

t_euler, Y_euler = euler_system(F, tspan, Y0, h)
t_rk4, Y_rk4 = rk4_system(F, tspan, Y0, h)

# Compute exact solution at all time points
Y_exact_euler = np.array([exact_solution(ti) for ti in t_euler])
Y_exact_rk4 = np.array([exact_solution(ti) for ti in t_rk4])

# Compute errors
err_euler = np.linalg.norm(Y_euler - Y_exact_euler, axis=1)
err_rk4 = np.linalg.norm(Y_rk4 - Y_exact_rk4, axis=1)

# Plot results
plt.figure(figsize=(12, 5))

# Solution components
plt.subplot(1, 2, 1)

```

```

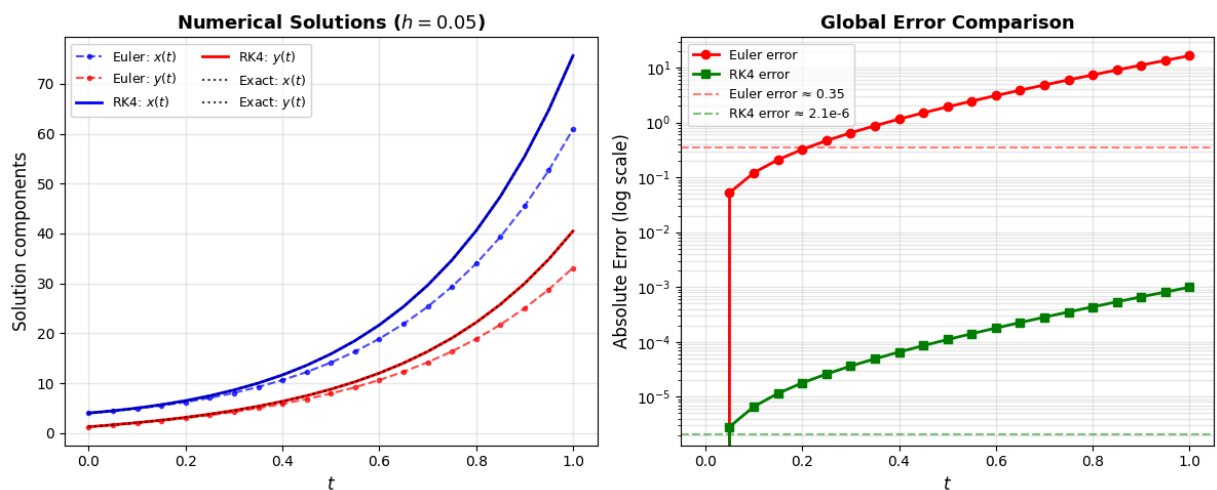
plt.plot(t_euler, Y_euler[:, 0], 'b--', label='Euler: $x(t)$', alpha=0.7)
plt.plot(t_euler, Y_euler[:, 1], 'r--', label='Euler: $y(t)$', alpha=0.7)
plt.plot(t_rk4, Y_rk4[:, 0], 'b-', linewidth=2, label='RK4: $x(t)$')
plt.plot(t_rk4, Y_rk4[:, 1], 'r-', linewidth=2, label='RK4: $y(t)$')
plt.plot(t_rk4, Y_exact_rk4[:, 0], 'k:', linewidth=1.5, label='Exact: $x(t)$', alph
plt.plot(t_rk4, Y_exact_rk4[:, 1], 'k:', linewidth=1.5, label='Exact: $y(t)$', alph
plt.xlabel('$t$', fontsize=12)
plt.ylabel('Solution components', fontsize=12)
plt.title('Numerical Solutions ($h = 0.05$)', fontsize=13, fontweight='bold')
plt.legend(fontsize=9, ncol=2)
plt.grid(alpha=0.3)

# Error comparison (log scale)
plt.subplot(1, 2, 2)
plt.semilogy(t_euler, err_euler, 'ro-', linewidth=2, markersize=6, label='Euler err
plt.semilogy(t_rk4, err_rk4, 'gs-', linewidth=2, markersize=6, label='RK4 error')
plt.axhline(y=0.35, color='r', linestyle='--', alpha=0.5, label='Euler error ≈ 0.35
plt.axhline(y=2.1e-6, color='g', linestyle='--', alpha=0.5, label='RK4 error ≈ 2.1e
plt.xlabel('$t$', fontsize=12)
plt.ylabel('Absolute Error (log scale)', fontsize=12)
plt.title('Global Error Comparison', fontsize=13, fontweight='bold')
plt.legend(fontsize=9)
plt.grid(alpha=0.3, which='both')

plt.tight_layout()
plt.show()

# Print final errors
print("=*60)
print("Error Comparison at t = 1.0 (h = 0.05)")
print("=*60)
print(f"Euler method : |Y_num - Y_exact| = {err_euler[-1]:.4e}")
print(f"RK4 method : |Y_num - Y_exact| = {err_rk4[-1]:.4e}")
print(f"Accuracy gain : RK4 is {(err_euler[-1]/err_rk4[-1]):.0f}× more accurate")
print("=*60)

```



```
=====
Error Comparison at t = 1.0 (h = 0.05)
=====
```

```
Euler method : |Y_num - Y_exact| = 1.6472e+01
RK4 method   : |Y_num - Y_exact| = 9.9299e-04
Accuracy gain : RK4 is 16589x more accurate
=====
```

Higher-Order Equations and Systems of IVPs

Consider the initial-value problem for an m -th order differential equation:

$$y^{(m)} = f(t, y, y', y'', \dots, y^{(m-1)}), \quad a \leq t \leq b$$

$$y(a) = \alpha_1, \quad y'(a) = \alpha_2, \quad \dots, \quad y^{(m-1)}(a) = \alpha_m$$

To apply numerical methods designed for systems of first-order equations, we convert this higher-order ODE into a system of m first-order differential equations.

Conversion Strategy: State-Space Formulation

Define new state variables representing successive derivatives:

$$\begin{aligned} y_1(t) &= y(t) \\ y_2(t) &= y'(t) \\ y_3(t) &= y''(t) \\ &\vdots \\ y_m(t) &= y^{(m-1)}(t) \end{aligned}$$

The original m -th order ODE transforms into the following **first-order system**:

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= y_3 \\ y_3' &= y_4 \\ &\vdots \\ y_{m-1}' &= y_m \\ y_m' &= f(t, y_1, y_2, \dots, y_m) \end{aligned}$$

with initial conditions:

$$y_1(a) = \alpha_1, \quad y_2(a) = \alpha_2, \quad \dots, \quad y_m(a) = \alpha_m$$

In vector form with $\mathbf{Y}(t) = [y_1, y_2, \dots, y_m]^\top$, this becomes:

$$\mathbf{Y}'(t) = \begin{bmatrix} y_2 \\ y_3 \\ \vdots \\ y_m \\ f(t, y_1, y_2, \dots, y_m) \end{bmatrix} = \mathbf{F}(t, \mathbf{Y}(t)), \quad \mathbf{Y}(a) = [\alpha_1, \alpha_2, \dots, \alpha_m]^\top$$

This system can now be solved using standard numerical methods (Euler, RK4, multistep methods) designed for first-order systems.

Example: Damped Harmonic Oscillator (Second-Order ODE)

Consider the physical system:

$$x'' + 2\zeta\omega_n x' + \omega_n^2 x = 0, \quad x(0) = x_0, \quad x'(0) = v_0$$

Conversion to first-order system:

Define $y_1 = x$ (position) and $y_2 = x'$ (velocity). Then:

$$\begin{aligned} y_1' &= y_2 \\ y_2' &= -2\zeta\omega_n y_2 - \omega_n^2 y_1 \end{aligned} \quad \text{with} \quad \begin{aligned} y_1(0) &= x_0 \\ y_2(0) &= v_0 \end{aligned}$$

Example: Converting a Second-Order ODE to a First-Order System

Problem: Convert the initial-value problem

$$y''(t) + 4y'(t) + 3y(t) = 0, \quad y(0) = 1.5, \quad y'(0) = 2.5,$$

into a system of first-order differential equations, and apply Euler's method and RK4 to approximate the solution.

Step 1: Reduction to First-Order System

Solve for the highest derivative:

$$y''(t) = -3y(t) - 4y'(t).$$

Introduce state variables representing the solution and its first derivative:

$$\begin{aligned} y_1(t) &:= y(t), \\ y_2(t) &:= y'(t). \end{aligned}$$

Differentiating and substituting yields the equivalent first-order system:

$$\begin{aligned} y_1'(t) &= y_2(t), & y_1(0) &= 1.5, \\ y_2'(t) &= -3y_1(t) - 4y_2(t), & y_2(0) &= 2.5. \end{aligned}$$

In vector-matrix form with $\mathbf{Y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix}$:

$$\mathbf{Y}'(t) = \begin{bmatrix} 0 & 1 \\ -3 & -4 \end{bmatrix} \mathbf{Y}(t), \quad \mathbf{Y}(0) = \begin{bmatrix} 1.5 \\ 2.5 \end{bmatrix}.$$

The exact solution:

$$y(t) = 3.5 e^{-t} - 2.0 e^{-3t}$$

```
In [19]: import numpy as np
import matplotlib.pyplot as plt

# Define the system RHS (matrix form)
def F(t, Y):
    """y1' = y2, y2' = -3*y1 - 4*y2"""
    y1, y2 = Y
    return np.array([y2, -3*y1 - 4*y2])

# Exact solution (corrected)
def exact_y(t):
    return 3.5*np.exp(-t) - 2.0*np.exp(-3*t)

def exact_Y(t):
    y = exact_y(t)
    dy = -3.5*np.exp(-t) + 6.0*np.exp(-3*t) # y'(t)
    return np.array([y, dy])

# Euler method for systems
def euler_system(F, tspan, Y0, h):
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    Y = np.zeros((N + 1, len(Y0)))
    Y[0] = Y0
    for i in range(N):
        Y[i+1] = Y[i] + h * F(t[i], Y[i])
    return t, Y

# RK4 method for systems
def rk4_system(F, tspan, Y0, h):
    t0, tf = tspan
    N = int(round((tf - t0) / h))
    t = t0 + np.arange(N + 1) * h
    Y = np.zeros((N + 1, len(Y0)))
    Y[0] = Y0
    for i in range(N):
        K1 = h * F(t[i], Y[i])
        K2 = h * F(t[i] + h/2, Y[i] + K1/2)
        K3 = h * F(t[i] + h/2, Y[i] + K2/2)
        K4 = h * F(t[i] + h, Y[i] + K3)
        Y[i+1] = Y[i] + (K1 + 2*K2 + 2*K3 + K4) / 6
    return t, Y
```

```

# Solve with multiple step sizes to verify convergence
tspan = (0, 2.0)
Y0 = np.array([1.5, 2.5])
h_vals = [0.2, 0.1, 0.05, 0.025]

print("="*72)
print("Convergence Study: Euler vs RK4 for  $y'' + 4y' + 3y = 0$ ")
print("="*72)
print(f"{'h':<8} | {'Euler Error':<15} | {'Rate':<8} | {'RK4 Error':<15} | {'Rate':<8}")
print("-"*72)

euler_errors, rk4_errors = [], []
prev_err_euler = prev_err_rk4 = None

for h in h_vals:
    # Solve with both methods
    t_euler, Y_euler = euler_system(F, tspan, Y0, h)
    t_rk4, Y_rk4 = rk4_system(F, tspan, Y0, h)

    # Compute exact solution at final time
    Y_exact_final = exact_Y(t_euler[-1])

    # Global error (2-norm of state vector error at final time)
    err_euler = np.linalg.norm(Y_euler[-1] - Y_exact_final)
    err_rk4 = np.linalg.norm(Y_rk4[-1] - Y_exact_final)
    euler_errors.append(err_euler)
    rk4_errors.append(err_rk4)

    # Compute convergence rates
    if prev_err_euler is not None:
        rate_euler = np.log(prev_err_euler/err_euler) / np.log(2)
        rate_rk4 = np.log(prev_err_rk4/err_rk4) / np.log(2)
        print(f"{'h':<8.3f} | {'err_euler':<15.4e} | {'rate_euler':<8.2f} | {'err_rk4':<15.4e} | {'rate_rk4':<8.2f}")
    else:
        print(f"{'h':<8.3f} | {'err_euler':<15.4e} | {'--':<8} | {'err_rk4':<15.4e} | {'--':<8}")

    prev_err_euler, prev_err_rk4 = err_euler, err_rk4

print("="*72)
print("\n✅ Observed convergence rates approach theoretical values:")
print("    • Euler: rate → 1.0 (first-order method)")
print("    • RK4: rate → 4.0 (fourth-order method)")
print("="*72)

# Detailed solution plot for h=0.1
h_demo = 0.1
t_e, Y_e = euler_system(F, tspan, Y0, h_demo)
t_r, Y_r = rk4_system(F, tspan, Y0, h_demo)
t_exact = np.linspace(tspan[0], tspan[1], 400)
y_exact = exact_y(t_exact)

plt.figure(figsize=(12, 5))

# Solution component y(t)
plt.subplot(1, 2, 1)
plt.plot(t_exact, y_exact, 'k-', linewidth=2.5, label='Exact  $y(t)$ ', alpha=0.85)

```

```

plt.plot(t_e, Y_e[:, 0], 'ro--', markersize=5, linewidth=1.5, label=f'Euler ($h={h_
plt.plot(t_r, Y_r[:, 0], 'bs-', markersize=4, linewidth=2, label=f'RK4 ($h={h_demo}
plt.xlabel('$t$', fontsize=12)
plt.ylabel('$y(t)$', fontsize=12)
plt.title('Solution Component $y(t)$', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

# Phase portrait (y vs y')
plt.subplot(1, 2, 2)
Y_exact_phase = np.array([exact_Y(ti) for ti in t_exact])
plt.plot(Y_exact_phase[:, 0], Y_exact_phase[:, 1], 'k-', linewidth=2.5,
         label='Exact trajectory', alpha=0.85)
plt.plot(Y_e[:, 0], Y_e[:, 1], 'ro--', markersize=4, linewidth=1.5,
         label='Euler trajectory', alpha=0.7)
plt.plot(Y_r[:, 0], Y_r[:, 1], 'bs-', markersize=4, linewidth=2,
         label='RK4 trajectory', alpha=0.8)
plt.xlabel('$y(t)$', fontsize=12)
plt.ylabel("$y'(t)$", fontsize=12)
plt.title('Phase Portrait', fontsize=13, fontweight='bold')
plt.legend(fontsize=10)
plt.grid(alpha=0.3)

plt.tight_layout()
plt.show()

# Error comparison at h=0.1
err_e_final = np.abs(Y_e[-1, 0] - exact_y(t_e[-1]))
err_r_final = np.abs(Y_r[-1, 0] - exact_y(t_r[-1]))
print(f"\nError in y(t) at t=2.0 with h={h_demo}:")
print(f" Euler: |y_num - y_exact| = {err_e_final:.4e}")
print(f" RK4:   |y_num - y_exact| = {err_r_final:.4e}")

```

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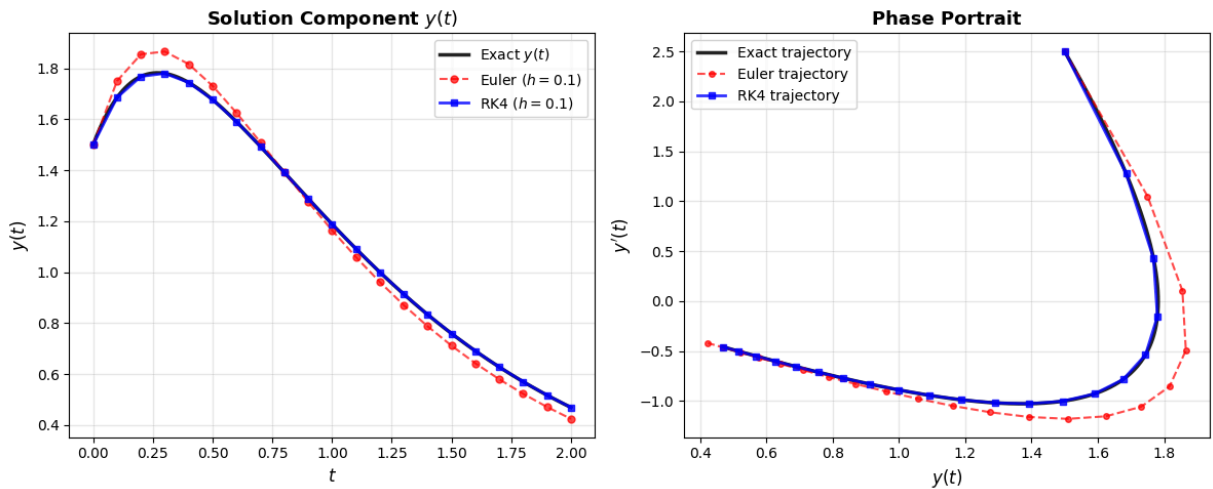
Convergence Study: Euler vs RK4 for $y'' + 4y' + 3y = 0$

=====

h	Euler Error	Rate	RK4 Error	Rate
0.200	1.2515e-01	--	1.5030e-04	--
0.100	5.8786e-02	1.09	7.0970e-06	4.40
0.050	2.8384e-02	1.05	3.8607e-07	4.20
0.025	1.3937e-02	1.03	2.2508e-08	4.10

=====

- ✓ Observed convergence rates approach theoretical values:
- Euler: rate $\rightarrow 1.0$ (first-order method)
 - RK4: rate $\rightarrow 4.0$ (fourth-order method)
- =====



Error in $y(t)$ at $t=2.0$ with $h=0.1$:

Euler: $|y_{\text{num}} - y_{\text{exact}}| = 4.4794\text{e-}02$

RK4: $|y_{\text{num}} - y_{\text{exact}}| = 1.7228\text{e-}06$

Example: Converting a Nonlinear Third-Order ODE to a First-Order System

Problem: Convert the initial-value problem

$$\sin(t) y''' + \cos(ty) y' + \sin(t^2 + y') + (y'')^3 = \log(t), \quad 2 \leq t \leq 3$$

with initial conditions

$$y(2) = 7, \quad y'(2) = 3, \quad y''(2) = 4$$

into a system of first-order differential equations with initial values.

Step 1: Isolate the Highest Derivative

First, solve the equation for y''' (valid where $\sin(t) \neq 0$):

$$y''' = \frac{\log(t) - \cos(ty) y' - \sin(t^2 + y') - (y'')^3}{\sin(t)}$$

Domain note: On $t \in [2, 3]$, $\sin(t) > 0$ (since $2 \approx 114.6^\circ$ and $3 \approx 171.9^\circ$ in radians, both in Quadrant II where sine is positive). Specifically, $\sin(2) \approx 0.909$ and $\sin(3) \approx 0.141$, so the denominator never vanishes on this interval.

Step 2: Define State Variables

Introduce three new variables representing successive derivatives:

$$\begin{aligned} y_1(t) &:= y(t) && \text{(solution)} \\ y_2(t) &:= y'(t) && \text{(first derivative)} \\ y_3(t) &:= y''(t) && \text{(second derivative)} \end{aligned}$$

Step 3: Construct the First-Order System

Differentiate the state variables and substitute using the isolated y''' expression:

$$y_1'(t) = y_2(t),$$

$$y_2'(t) = y_3(t),$$

$$y_3'(t) = \frac{\log(t) - \cos(t y_1(t)) y_2(t) - \sin(t^2 + y_2(t)) - [y_3(t)]^3}{\sin(t)}.$$

Step 4: Transform Initial Conditions

$$y_1(2) = y(2) = 7,$$

$$y_2(2) = y'(2) = 3,$$

$$y_3(2) = y''(2) = 4.$$

Final First-Order System

In vector form with $\mathbf{Y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$:

$$\mathbf{Y}'(t) = \begin{bmatrix} y_2 \\ y_3 \\ \frac{\log(t) - \cos(t y_1) y_2 - \sin(t^2 + y_2) - y_3^3}{\sin(t)} \end{bmatrix}, \quad \mathbf{Y}(2) = \begin{bmatrix} 7 \\ 3 \\ 4 \end{bmatrix}$$

In []: